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2 Quantum Fields

Exercise 2.1

The partition function is

$$\begin{aligned} Z &= \sum_n e^{-\beta \hbar \omega (n + \frac{1}{2})} \\ &= e^{-\beta \hbar \omega / 2} \frac{1}{1 - e^{-\beta \hbar \omega}}. \end{aligned} \quad (2.1.1)$$

Let $x = \beta \hbar \omega$,

$$Z = e^{-x/2} (1 - e^{-x})^{-1}. \quad (2.1.2)$$

The internal energy is

$$\begin{aligned} E &= N_A \left(-\frac{\partial \ln Z}{\partial \beta} \right) \\ &= -N_A \hbar \omega \frac{1}{Z} \frac{\partial Z}{\partial x} \\ &= \frac{1}{2} N_A \hbar \omega \left(1 + \frac{e^{-x/2}}{\sinh \frac{x}{2}} \right). \end{aligned} \quad (2.1.3)$$

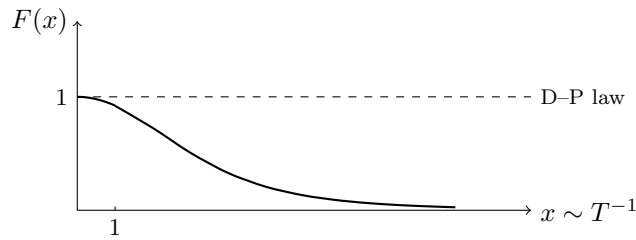
The heat capacity is

$$\begin{aligned} C_V &= \frac{\partial E}{\partial T} = \frac{dx}{dT} \frac{\partial E}{\partial x} \\ &= \frac{1}{4} R x^2 \frac{1}{\sinh^2(\frac{x}{2})} \\ &= R F(x) \end{aligned} \quad (2.1.4)$$

where

$$F(x) = \left(\frac{x/2}{\sinh(x/2)} \right)^2 \rightarrow 1 \quad (2.1.5)$$

in the $x \rightarrow 0$ limit ($k_B T \gg \hbar \omega$), where it recovers the Dulong–Petit law.



Exercise 2.2

$$\begin{aligned} \sum_j \langle q_m | j \rangle \langle j | q_n \rangle &= \frac{1}{N_S} \sum_{j=0}^{N_S-1} e^{i(q_n - q_m)ja} \\ &= \frac{1}{N_S} \sum_{j=0}^{N_S-1} e^{i(n-m)\frac{2\pi}{N_S}j}. \end{aligned} \quad (2.2.1)$$

If $n \neq m$, then¹

$$\begin{aligned}
 \text{LHS} &= \frac{1}{N_S} \frac{e^{i2\pi(n-m)} - 1}{e^{i2\pi(n-m)/N_S} - 1} \\
 &= \frac{1}{N_S} \frac{\sin(\pi(n-m))}{\sin\left(\frac{\pi}{N_S}(n-m)\right)} \frac{e^{-i\pi(n-m)/N_S}}{e^{-i\pi(n-m)}} \\
 &= \frac{1}{N_S} \frac{\sin(\pi(n-m))}{\sin\left(\frac{\pi}{N_S}(n-m)\right)} (-1)^{\frac{N_S-1}{N_S}(n-m)} \\
 &= 0.
 \end{aligned} \tag{2.2.2}$$

If $n = m$, then

$$\begin{aligned}
 \text{LHS} &= \frac{1}{N_S} \sum_{j=0}^{N_S-1} 1 \\
 &= 1.
 \end{aligned} \tag{2.2.3}$$

Therefore,

$$\langle q_m | q_n \rangle = \delta_{mn}. \tag{2.2.4}$$

Exercise 2.3

Have

$$\hat{x}(0) = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \tag{2.3.1}$$

$$\hat{x}(t) = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} e^{i\omega t} + \hat{a}^\dagger e^{-i\omega t}). \tag{2.3.2}$$

The symmetrised correlation function can be expanded as

$$\begin{aligned}
 S_\beta(t) &= \frac{1}{2} \langle \hat{x}(t) \hat{x}(0) + \hat{x}(0) \hat{x}(t) \rangle \\
 &= \frac{\hbar}{4m\omega} [2e^{i\omega t} \langle \hat{a} \hat{a} \rangle + 2e^{-i\omega t} \langle \hat{a}^\dagger \hat{a}^\dagger \rangle + 2 \cos \omega t (\langle \hat{a} \hat{a}^\dagger \rangle + \langle \hat{a}^\dagger \hat{a} \rangle)].
 \end{aligned} \tag{2.3.3}$$

At (inverse) temperature β ,

$$\begin{cases} \langle \hat{a}^\dagger \hat{a} \rangle = \langle \hat{n} \rangle \\ \langle \hat{a} \hat{a}^\dagger \rangle = \langle \hat{a}^\dagger \hat{a} + 1 \rangle = \langle \hat{a} \rangle + 1 \\ \langle \hat{a} \hat{a} \rangle = \langle \hat{a}^\dagger \hat{a}^\dagger \rangle = 0, \end{cases} \tag{2.3.4}$$

where

$$\langle \hat{n} \rangle = n(\beta) = \frac{1}{e^{\beta \hbar \omega} - 1}. \tag{2.3.5}$$

Substituting in (2.3.3) gives

$$S_\beta(t) = \frac{\hbar}{2m\omega} \cos(\omega t) \left[\underbrace{2n(\beta)}_{\text{finite-}T \text{ excitation}} + \underbrace{1}_{\text{zero-point fluctuation}} \right]. \tag{2.3.6}$$

¹I believe the stated result is missing the red N_S^{-1} .

Exercise 2.4

(a)

$$[\hat{b}, \hat{b}] = u^2[\hat{a}, \hat{a}] + uv[\hat{a}, \hat{a}^\dagger] + uv[\hat{a}^\dagger, \hat{a}] + v^2[\hat{a}^\dagger, \hat{a}^\dagger] = 0 \quad (2.4.1)$$

$$[\hat{b}^\dagger, \hat{b}^\dagger] = u^2[\hat{a}^\dagger, \hat{a}^\dagger] + uv[\hat{a}^\dagger, \hat{a}] + uv[\hat{a}, \hat{a}^\dagger] + v^2[\hat{a}, \hat{a}] = 0 \quad (2.4.2)$$

$$[\hat{b}, \hat{b}^\dagger] = u^2[\hat{a}, \hat{a}^\dagger] + uv[\hat{a}, \hat{a}] + uv[\hat{a}^\dagger, \hat{a}^\dagger] + v^2[\hat{a}^\dagger, \hat{a}] = u^2 - v^2. \quad (2.4.3)$$

The first two canonical commutation relations of $\hat{b}$ are guaranteed by the canonical commutation relations of $\hat{a}$, while the last one satisfies $[\hat{b}, \hat{b}^\dagger] = 1$ if and only if $u^2 - v^2 = 1$.

This means that it is convenient to parameterise u, v as $u = \cosh \theta$, $v = \sinh \theta$.

(b) Given $u^2 - v^2 = 1$, the inverse transform is

$$\hat{a} = u\hat{b} - v\hat{b}^\dagger\hat{a}^\dagger = u\hat{b}^\dagger - v\hat{b}. \quad (2.4.4)$$

Then we can expand

$$\begin{aligned} \hat{a}^\dagger\hat{a} + \frac{1}{2} &= u^2\hat{b}^\dagger\hat{b} - uv\hat{b}^\dagger\hat{b}^\dagger - uv\hat{b}\hat{b} + v^2\hat{b}^\dagger\hat{b}^\dagger + \frac{1}{2} \\ &= (u^2 + v^2) \left(\hat{b}^\dagger\hat{b} + \frac{1}{2} \right) - uv(\hat{b}^\dagger\hat{b}^\dagger + \hat{b}\hat{b}) \end{aligned} \quad (2.4.5)$$

and similarly

$$\hat{a}\hat{a} + \hat{a}^\dagger\hat{a}^\dagger = (u^2 + v^2)(\hat{b}\hat{b} + \hat{b}^\dagger\hat{b}^\dagger) - 4uv \left(\hat{b}^\dagger\hat{b} + \frac{1}{2} \right) \quad (2.4.6)$$

using $u^2 - v^2 = 1$. Therefore,

$$\hat{H} = [\omega(u^2 + v^2) - 2\Delta uv] \left(\hat{b}^\dagger\hat{b} + \frac{1}{2} \right) + \left[\frac{\Delta}{2}(u^2 + v^2) - \omega uv \right] (\hat{b}^\dagger\hat{b}^\dagger + \hat{b}\hat{b}). \quad (2.4.7)$$

For $\hat{H}$ to be diagonalisable, there must be some θ such that for $u = \cosh \theta$, $v = \sinh \theta$, the latter term vanish,

$$\Delta(u^2 + v^2) = 2\omega uv. \quad (2.4.8)$$

In θ , this is

$$\Delta \cosh 2\theta = \omega \sinh 2\theta, \quad (2.4.9)$$

so

$$\tanh 2\theta = \frac{\Delta}{\omega}. \quad (2.4.10)$$

Therefore, we must have

$$|\Delta| < \omega \quad (2.4.11)$$

for $\hat{H}$ to be diagonalisable.

The new frequency is

$$\begin{aligned} \tilde{\omega} &= \omega(u^2 + v^2) - 2\Delta uv \\ &= \omega \cosh 2\theta - \Delta \sinh 2\theta. \end{aligned} \quad (2.4.12)$$

Given $\tanh 2\theta = \frac{\Delta}{\omega}$, we have

$$\begin{cases} \sinh 2\theta = \frac{\Delta}{\sqrt{\omega^2 - \Delta^2}} \\ \cosh 2\theta = \frac{\omega}{\sqrt{\omega^2 - \Delta^2}}, \end{cases} \quad (2.4.13)$$

so

$$\tilde{\omega} = \sqrt{\omega^2 - \Delta^2}. \quad (2.4.14)$$

Since

$$\begin{cases} u^2 + v^2 = \cosh 2\theta = \frac{\omega}{\tilde{\omega}} \\ u^2 - v^2 = 1, \end{cases} \quad (2.4.15)$$

we have

$$\begin{cases} u^2 = \frac{1}{2} \left(\frac{\omega}{\tilde{\omega}} + 1 \right) \\ v^2 = \frac{1}{2} \left(\frac{\omega}{\tilde{\omega}} - 1 \right). \end{cases} \quad (2.4.16)$$

We need to take care of the signs when taking the square roots.

$$\begin{cases} u = \sqrt{\frac{1}{2} \left(\frac{\omega}{\sqrt{\omega^2 - \Delta^2}} + 1 \right)} \\ v = \operatorname{sgn}(\Delta) \sqrt{\frac{1}{2} \left(\frac{\omega}{\sqrt{\omega^2 - \Delta^2}} - 1 \right)}. \end{cases} \quad (2.4.17)$$

When $\Delta = \omega$, the transition is singular. As $\theta \rightarrow \infty$, $u, v \rightarrow \infty$ but $\tilde{\omega} \rightarrow 0$. However, this does not mean that $\hat{H}$ is 0 (as it does not, apparently). In fact,

$$\begin{aligned} \hat{H} &= \frac{1}{2} \omega (\hat{a} + \hat{a}^\dagger)^2 \\ &= \omega \hat{x}^2. \end{aligned} \quad (2.4.18)$$

The momentum term vanishes.

(c) We have

$$e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger} = \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha \hat{a}^\dagger \hat{a}^\dagger)^k. \quad (2.4.19)$$

The commutation relations are

$$[\hat{a}^\dagger, e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger}] = 0 \quad (2.4.20)$$

and

$$\begin{aligned} [\hat{a}, e^{-\alpha \hat{a}^\dagger \hat{a}^\dagger}] &= \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha)^k [\hat{a}, (\hat{a}^\dagger)^{2k}] \\ &= \sum_{k=1}^{\infty} \frac{1}{k!} (-\alpha)^k \cdot 2k (\hat{a}^\dagger)^{2k-1}, \end{aligned} \quad (2.4.21)$$

where in evaluating $[\hat{a}, (\hat{a}^\dagger)^n]$, notice that when you move $\hat{a}$ to the right of a string of $\hat{a}^\dagger$'s, each time you move through an $\hat{a}^\dagger$, you get a $[\hat{a}, \hat{a}^\dagger] = 1$ multiplied with the other $(n-1)$ $\hat{a}^\dagger$'s.

Therefore,

$$\begin{aligned}
\hat{b}|\tilde{0}\rangle &= (u\hat{a} + v\hat{a}^\dagger)e^{-\alpha\hat{a}^\dagger\hat{a}}|0\rangle \\
&= \left[u \left(e^{-\alpha\hat{a}^\dagger\hat{a}}\hat{a} + [\hat{a}, e^{-\alpha\hat{a}^\dagger\hat{a}}] \right) + v e^{-\alpha\hat{a}^\dagger\hat{a}}\hat{a}^\dagger \right] |0\rangle \\
&= \left\{ u \sum_{k=1}^{\infty} \frac{1}{k!} (-\alpha)^k 2k (\hat{a}^\dagger)^{2k-1} + v \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha)^k (\hat{a}^\dagger)^{2k+1} \right\} |0\rangle \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} (-\alpha)^k \left[-\frac{\alpha}{k+1} \cdot 2(k+1)u + v \right] (\hat{a}^\dagger)^{2k+1} |0\rangle.
\end{aligned} \tag{2.4.22}$$

For this to be 0, each term in the sum must be zero, and hence

$$\alpha = \frac{v}{2u}. \tag{2.4.23}$$

Alternatively: suppose

$$|\tilde{0}\rangle = \sum_{n=0}^{\infty} c_n |n\rangle. \tag{2.4.24}$$

Then

$$\begin{aligned}
\hat{b}|\tilde{0}\rangle &= (u\hat{a} + v\hat{a}^\dagger) \sum_{n=0}^{\infty} c_n |n\rangle \\
&= u \sum_{n=1}^{\infty} c_n \sqrt{n} |n-1\rangle + v \sum_{n=0}^{\infty} c_n \sqrt{n+1} |n+1\rangle \\
&= uc_1 |0\rangle + \sum_{n=1}^{\infty} (uc_{n+1}\sqrt{n+1} + vc_{n-1}\sqrt{n}) |n\rangle.
\end{aligned} \tag{2.4.25}$$

Each term must vanish, so

$$\begin{cases} uc_1 = 0 \\ uc_{n+1}\sqrt{n+1} + vc_{n-1}\sqrt{n} = 0 \quad \text{for } n \geq 1. \end{cases} \tag{2.4.26}$$

The first condition implies $c_1 = 0$, while the second condition gives the iterative formula for $n \geq 2$:

$$c_n = -\frac{v}{u} \sqrt{\frac{n-1}{n}} c_{n-2}. \tag{2.4.27}$$

This means that all the odd terms must vanish:

$$c_{2k+1} = 0 \tag{2.4.28}$$

for $k \in \mathbb{N}$, while the even coefficients are

$$c_{2k} = \left(-\frac{v}{u}\right)^k \sqrt{\frac{(2k-1)!!}{(2k)!!}} c_0. \tag{2.4.29}$$

Therefore,

$$\begin{aligned}
|\tilde{0}\rangle &= \sum_{k=0}^{\infty} c_{2k} |2k\rangle \\
&= \sum_{k=0}^{2k} \left(-\frac{v}{u}\right)^k \sqrt{\frac{(2k-1)!!}{(2k)!!}} \frac{(\hat{a}^\dagger)^{2k}}{(2k)!} |0\rangle \\
&= \sum_{k=0}^{\infty} \left(-\frac{v}{u}\right)^k \frac{1}{(2k)!!} (\hat{a}^\dagger)^{2k} |0\rangle \\
&= \left\{ \sum_{k=0}^{\infty} \frac{1}{k!} \left[-\frac{v}{2u} (\hat{a}^\dagger)^2\right]^k \right\} |0\rangle \\
&= \exp\left(-\frac{v}{2u} \hat{a}^\dagger \hat{a}^\dagger\right) |0\rangle.
\end{aligned} \tag{2.4.30}$$

Note that the new vacuum is not the old vacuum. It contains indefinite, even number of the original bosons.

Exercise 2.5

(a) Let $u_\ell = \phi_{2k}$, $v_\ell = \phi_{2k+1}$. The equation of motion is

$$\begin{cases} m\ddot{u}_\ell = -k(2u_\ell - v_{\ell-1} - v_\ell) \\ M\ddot{v}_\ell = -k(2v_\ell - u_\ell - u_{\ell+1}). \end{cases} \tag{2.5.1}$$

Transforming into the Fourier basis

$$\begin{cases} U_q = \frac{1}{\sqrt{N}} \sum_\ell e^{-2i\ell a q} u_\ell \\ V_q = \frac{1}{\sqrt{N}} \sum_\ell e^{-2i\ell a q} v_\ell, \end{cases} \tag{2.5.2}$$

where q lives in the first Brillouin zone $q \in (-\frac{\pi}{2a}, \frac{\pi}{2a}]$, the equation of motion is

$$\begin{cases} m\ddot{U}_q = -2kU_q + 2k \cos(aq)V_q \\ M\ddot{V}_q = -2kV_q + 2k \cos(aq)U_q. \end{cases} \tag{2.5.3}$$

In matrix notation,

$$\begin{pmatrix} m & 0 \\ 0 & M \end{pmatrix} \begin{pmatrix} \ddot{U}_q \\ \ddot{V}_q \end{pmatrix} = - \begin{pmatrix} 2k & -2k \cos(qa) \\ -2k \cos(qa) & 2k \end{pmatrix} \begin{pmatrix} U_q \\ V_q \end{pmatrix}, \tag{2.5.4}$$

or

$$\mathbf{M}\ddot{\mathbf{U}}_q = -\mathbf{K}_q \mathbf{U}_q. \tag{2.5.5}$$

Using the ansatz $\mathbf{U}_q(t) = \mathbf{A}_q e^{i\omega_q t}$, this is

$$\omega_q^2 \mathbf{M} \mathbf{A}_q = \mathbf{K}_q \mathbf{A}_q. \tag{2.5.6}$$

This is a generalised eigenvalue equation, which is not the easiest to solve. This inspires us to introduce the mass-scaled coordinates

$$\mathbf{X}_q = \begin{pmatrix} X_q \\ Y_q \end{pmatrix} = \begin{pmatrix} \sqrt{m} U_q \\ \sqrt{M} V_q \end{pmatrix} \tag{2.5.7}$$

such that

$$\mathbf{U}_q = \mathbf{M}^{-1/2} \mathbf{X}_q. \tag{2.5.8}$$

Then we have

$$\ddot{\mathbf{X}} = -\mathbf{D}_q \mathbf{X}_q \quad (2.5.9)$$

where

$$\mathbf{D} = \mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2} = \begin{pmatrix} \frac{2k}{m} & -\frac{2k \cos(qa)}{\sqrt{mM}} \\ -\frac{2k \cos(qa)}{\sqrt{mM}} & \frac{2k}{M} \end{pmatrix}. \quad (2.5.10)$$

Now suppose $\mathbf{X}_q(t) = \mathbf{A}_q e^{i\omega_q t}$, we have the eigenvalue equation

$$\mathbf{D}_q \mathbf{A}_q = \omega_q^2 \mathbf{A}_q. \quad (2.5.11)$$

Solving for the eigenvalue equation

$$\det(\mathbf{D}_q - \omega_q^2 \mathbf{I}) = 0 \quad (2.5.12)$$

gives the two eigenvalues

$$\omega_{\pm}^2(q) = k \left(\frac{1}{m} + \frac{1}{M} \right) \pm k \sqrt{\left(\frac{1}{m} + \frac{1}{M} \right)^2 - \frac{4}{mM} \sin^2(qa)}. \quad (2.5.13)$$

We denote the two normalised eigenvectors

$$\mathbf{e}_{\pm}(q) = \begin{pmatrix} e_{\pm,1} \\ e_{\pm,2} \end{pmatrix}. \quad (2.5.14)$$

They satisfy

$$\left(\frac{2k}{m} - \omega_{\pm}^2 \right) e_{\pm,1} - \frac{2k \cos(qa)}{\sqrt{mM}} e_{\pm,2} = 0, \quad (2.5.15)$$

which gives

$$\frac{e_{\pm,2}}{e_{\pm,1}} = \frac{\sqrt{mM} \left(\frac{2k}{m} - \omega_{\pm}^2 \right)}{2k \cos(qa)}. \quad (2.5.16)$$

This should be solved along with the normalisation condition

$$e_{\pm,1}^2 + e_{\pm,2}^2 = 1, \quad (2.5.17)$$

but I do not bother to.

Let

$$\mathbf{E}_q = \begin{pmatrix} | & | \\ \mathbf{e}_+(q) & \mathbf{e}_-(q) \\ | & | \end{pmatrix}, \quad (2.5.18)$$

then

$$\mathbf{E}_q^T \mathbf{D}_q \mathbf{E}_q = \begin{pmatrix} \omega_+^2(q) & 0 \\ 0 & \omega_-^2(q) \end{pmatrix} =: \Omega_q, \quad (2.5.19)$$

and so

$$\ddot{\mathbf{X}}_q = -\mathbf{E}_q \Omega_q \mathbf{E}_q^T \mathbf{X}_q. \quad (2.5.20)$$

Therefore

$$\frac{d^2}{dt^2} (\mathbf{E}_q^T \mathbf{X}_q) = -\Omega_q (\mathbf{E}_q^T \mathbf{X}_q). \quad (2.5.21)$$

This is a diagonal equation of motion, and

$$\mathbf{Q}_q = \mathbf{E}_q^T \mathbf{X}_q \quad (2.5.22)$$

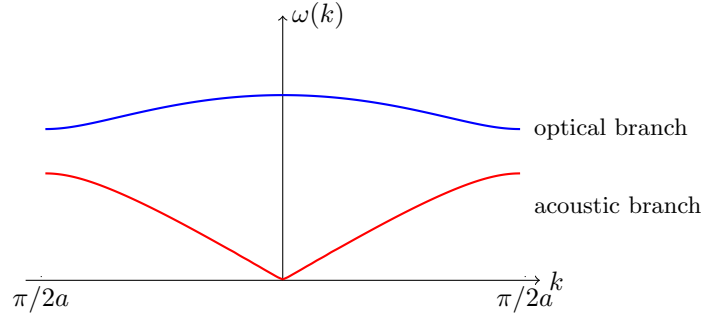
are the normal coordinates. The two normal coordinates are

$$Q_{q,\pm} = \mathbf{e}_{q,\pm}^T \mathbf{X}_q. \quad (2.5.23)$$

They satisfy

$$\ddot{Q}_{q,\pm} = -\omega_{q,\pm}^2 Q_{q,\pm}, \quad (2.5.24)$$

so that each is oscillating independently with frequency $\omega_{q,\pm}$.



(b) At the band edge $q = \frac{\pi}{2a}$,

$$\omega_{\pm}^2 = \frac{k}{mM} [(m+M) \pm (m-M)], \quad (2.5.25)$$

so

$$\begin{cases} \omega_+ = \sqrt{\frac{2k}{M}} \\ \omega_- = \sqrt{\frac{2k}{m}} \end{cases} . \quad (2.5.26)$$

The band gap is

$$\Delta\omega = \sqrt{2k} \left| \frac{1}{\sqrt{M}} - \frac{1}{\sqrt{m}} \right|. \quad (2.5.27)$$

(c) We will denote the momentum conjugate to $\mathbf{X}_q$ as $\mathbf{P}$, and that to $\mathbf{Q}_q$ as $\mathbf{\Pi}_q$. The kinetic energy is

$$\begin{aligned} T &= \frac{1}{2} \sum_q m \left| \dot{\mathbf{U}}_q \right|^2 + M \left| \dot{\mathbf{V}}_q \right|^2 \\ &= \frac{1}{2} \sum_q \dot{\mathbf{X}}_q^\dagger \dot{\mathbf{X}}_q \\ &= \frac{1}{2} \sum_q \mathbf{P}_q^\dagger \mathbf{P}_q \\ &= \frac{1}{2} \sum_q \mathbf{\Pi}_q^\dagger \mathbf{\Pi}_q, \end{aligned} \quad (2.5.28)$$

and the potential energy is

$$\begin{aligned} V &= \frac{1}{2} \sum_q \mathbf{X}_q^\dagger \mathbf{D}_q \mathbf{X}_q \\ &= \frac{1}{2} \sum_q \mathbf{Q}_q^\dagger \Omega_q \mathbf{Q}_q. \end{aligned} \quad (2.5.29)$$

The Hamiltonian is therefore

$$H = \sum_q \sum_{\sigma=\pm} \left[\frac{1}{2} |\Pi_{q,\sigma}|^2 + \frac{1}{2} \omega_{q,\sigma}^2 |Q_{q,\sigma}|^2 \right]. \quad (2.5.30)$$

Quantising:

$$\hat{a}_{q,\sigma} = \sqrt{\frac{\omega_{q,\sigma}}{2\hbar}} Q_{q,\sigma} + \frac{i}{\sqrt{2\hbar\omega_{q,\sigma}}} \Pi_{q,\sigma} \quad (2.5.31)$$

such that

$$[\hat{a}_{q,\sigma}, \hat{a}_{q',\sigma'}^\dagger] = \delta_{qq'} \delta_{\sigma\sigma'}. \quad (2.5.32)$$

The Hamiltonian is

$$\hat{H} = \sum_q \sum_{\sigma=\pm} \left[\hbar\omega_{q,\sigma} \left(\hat{a}_{q,\sigma}^\dagger \hat{a}_{q,\sigma} + \frac{1}{2} \right) \right]. \quad (2.5.33)$$

Exercise 2.6

(a) The RMS displacement

$$\Delta r_{\text{RMS}} = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}, \quad (2.6.1)$$

where in each direction,

$$\Delta x \sim \sqrt{\frac{\hbar}{m\omega}} \quad (2.6.2)$$

by the uncertainty principle. Therefore, by Linderman criterion, the crystal is unstable if

$$\Delta r_{\text{RMS}} \sim \sqrt{\frac{\hbar}{m\omega}} \gtrsim \frac{1}{3}a \implies \frac{\hbar}{m\omega a^2} > \zeta_c \quad (2.6.3)$$

for some dimensionless ζ_c .

(b) Have

$$\begin{aligned} \hat{\Phi}_j &= \frac{1}{\sqrt{N_S}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}_j} \hat{\Phi}_{\mathbf{q}} \\ &= \sqrt{\frac{\hbar}{2mN_S}} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}_j} \frac{1}{\sqrt{\omega_{\mathbf{q}}}} (\hat{a}_{\mathbf{q}} + \hat{a}_{-\mathbf{q}}^\dagger). \end{aligned} \quad (2.6.4)$$

Take $j = 0$,

$$\hat{\Phi} = \sqrt{\frac{\hbar}{2mN_S}} \sum_{\mathbf{q}} \frac{1}{\sqrt{\omega_{\mathbf{q}}}} (\hat{a}_{\mathbf{q}} + \hat{a}_{-\mathbf{q}}^\dagger). \quad (2.6.5)$$

To work out $\omega_{\mathbf{q}}$, consider

$$\Phi_{\mathbf{R}+\mathbf{a}} - \Phi_{\mathbf{R}} = \frac{1}{\sqrt{N_S}} \sum_{\mathbf{q}} \Phi_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{R}} (e^{i\mathbf{q} \cdot \mathbf{a}} + 1). \quad (2.6.6)$$

By Fourier orthogonality

$$\sum_{\mathbf{R}} e^{i(\mathbf{q}+\mathbf{q}') \cdot \mathbf{R}} = N \delta_{\mathbf{q}', -\mathbf{q}}, \quad (2.6.7)$$

we have

$$\begin{aligned} \sum_{\mathbf{R}} (\Phi_{\mathbf{R}+\mathbf{a}} - \Phi_{\mathbf{R}})^2 &= \sum_{\mathbf{q}} \Phi_{\mathbf{q}} \Phi_{-\mathbf{q}} (e^{i\mathbf{q} \cdot \mathbf{a}} - 1)(e^{-i\mathbf{q} \cdot \mathbf{a}} - 1) \\ &= \sum_{\mathbf{q}} 4 \sin^2 \left(\frac{\mathbf{q} \cdot \mathbf{a}}{2} \right) \Phi_{\mathbf{q}}^\dagger \Phi_{\mathbf{q}}. \end{aligned} \quad (2.6.8)$$

Summing over the three directions $\mathbf{a} \in \{a\hat{\mathbf{x}}, a\hat{\mathbf{y}}, a\hat{\mathbf{z}}\}$,

$$V = \frac{1}{2} m \omega^2 \sum_{\mathbf{q}} 4 \sin \left[\sin^2 \frac{q_x a}{2} + \sin^2 \frac{q_y a}{2} + \sin^2 \frac{q_z a}{2} \right] \Phi_{\mathbf{q}}^\dagger \Phi_{\mathbf{q}}, \quad (2.6.9)$$

and hence

$$\omega_{\mathbf{q}} = 2\omega \sqrt{\sin^2 \frac{q_x a}{2} + \sin^2 \frac{q_y a}{2} + \sin^2 \frac{q_z a}{2}}. \quad (2.6.10)$$

Therefore,

$$\begin{aligned}
\langle 0 | \hat{\Phi}^2 | 0 \rangle &= \frac{\hbar}{2mN_S} \sum_{\mathbf{q}, \mathbf{q}'} \frac{1}{\sqrt{\omega_{\mathbf{q}} \omega_{\mathbf{q}'}}} \langle 0 | (\hat{a}_{\mathbf{q}} + \hat{a}_{-\mathbf{q}}^\dagger)(\hat{a}_{\mathbf{q}'} + \hat{a}_{-\mathbf{q}'}^\dagger) | 0 \rangle \\
&= \frac{\hbar}{2mN_S} \sum_{\mathbf{q}, \mathbf{q}'} \frac{1}{\sqrt{\omega_{\mathbf{q}} \omega_{\mathbf{q}'}}} \underbrace{\langle 0 | \hat{a}_{\mathbf{q}} \hat{a}_{-\mathbf{q}'}^\dagger | 0 \rangle}_{\delta_{\mathbf{q}, -\mathbf{q}'}} \\
&= \frac{\hbar}{2mN_S} \sum_{\mathbf{q}} \frac{1}{\omega_{\mathbf{q}}} .
\end{aligned} \tag{2.6.11}$$

We will approximate the sum by an integral

$$\frac{1}{N_S} \sum_{\mathbf{q}} \longrightarrow \left(\frac{a}{2\pi} \right)^3 \int_{-\pi/a}^{\pi/a} d^3 \mathbf{q} . \tag{2.6.12}$$

Then

$$\begin{aligned}
\langle 0 | \hat{\Phi}^2 | 0 \rangle &= \frac{\hbar}{4\pi\omega} \int_{-\pi}^{\pi} \frac{-\pi}{\pi} \frac{d^3 \mathbf{x}}{(2\pi)^3} \frac{1}{\sqrt{\sin^2(\frac{x}{2}) + \sin^2(\frac{y}{2}) + \sin^2(\frac{z}{2})}} \\
&= \frac{\hbar}{4m\omega} I ,
\end{aligned} \tag{2.6.13}$$

where

$$I = 0.910688 \dots \tag{2.6.14}$$

For the crystal to be unstable,

$$\langle 0 | \hat{\Phi}^2 | 0 \rangle > \frac{\hbar}{4m\omega} I > \frac{1}{9} a^2 , \tag{2.6.15}$$

and so

$$\frac{\hbar}{m\omega a^2} > \frac{4}{9I} =: \zeta_c \approx 0.4880 \dots \tag{2.6.16}$$

(c) Under the unrealistic assumption that ω is independent of a , Helium is *stable* if

$$\frac{\hbar}{m\omega a^2} < \zeta_c , \tag{2.6.17}$$

giving

$$a > a_c = \sqrt{\frac{\hbar}{m\omega \zeta_c}} \approx 2.878 \times 10^{-11} \text{ m} . \tag{2.6.18}$$

This is apparently unrealistic since crystalline Helium should be *unstable* at low density (for a greater than some a_c).

Using the realistic assumption that $\omega \sim a^{-\alpha}$ for some $\alpha > 2$, we now have that Helium is *stable* if

$$\frac{\hbar}{m\omega a^2} = \frac{\hbar}{m\omega_0} a^\gamma < \zeta_c \tag{2.6.19}$$

for some $\gamma = \alpha - 2 > 0$. This is

$$a < \left(\frac{\zeta_c m \omega_0}{\hbar} \right)^{1/\gamma} =: a_c . \tag{2.6.20}$$

Exercise 2.7

(a) Transforming into the Fourier basis,

$$\begin{aligned}
 \sum_j \hat{a}_{j+1}^\dagger \hat{a}_j &= \sum_j \sum_q \sum_p e^{-i(q-p)ja} e^{-iqa} \hat{a}_q^\dagger \hat{a}_p \\
 &= \sum_q \sum_p \delta_{pq} e^{-iqa} \hat{a}_q^\dagger \hat{a}_p \\
 &= \sum_q e^{-iqa} \hat{a}_q^\dagger \hat{a}_q,
 \end{aligned} \tag{2.7.1}$$

and its Hermitian conjugate is

$$\sum_q e^{+iqa} \hat{a}_q^\dagger \hat{a}_q. \tag{2.7.2}$$

Therefore, the first term in the Hamiltonian is

$$\hat{H}_q = J_1 \sum_q 2 \cos(qa) \hat{a}_q^\dagger \hat{a}_q. \tag{2.7.3}$$

Next,

$$\begin{aligned}
 \sum_j \hat{a}_{j+1}^\dagger \hat{a}_j^\dagger &= \sum_j \sum_q \sum_p e^{-i(q+p)ja} e^{-iqa} \hat{a}_q^\dagger \hat{a}_p^\dagger \\
 &= \sum_q \sum_p \delta_{p(-q)} e^{-iqa} \hat{a}_q^\dagger \hat{a}_p^\dagger \\
 &= \sum_q e^{-iqa} \hat{a}_q^\dagger \hat{a}_{-q}^\dagger,
 \end{aligned} \tag{2.7.4}$$

Since $\hat{a}_q^\dagger$ commutes with $\hat{a}_{-q}^\dagger$, we can set $q \rightarrow -q$ and write

$$\sum_j \hat{a}_{j+1}^\dagger \hat{a}_j^\dagger = \sum_q e^{+iqa} \hat{a}_q^\dagger \hat{a}_{-q}^\dagger. \tag{2.7.5}$$

Taking the average of the two expressions,

$$\sum_j \hat{a}_{j+1}^\dagger \hat{a}_j^\dagger = \sum_q \cos(qa) \hat{a}_q^\dagger \hat{a}_{-q}^\dagger, \tag{2.7.6}$$

and its Hermitian conjugate is

$$\sum_q \cos(qa) \hat{a}_q \hat{a}_{-q}, \tag{2.7.7}$$

so the second part of the Hamiltonian is

$$\hat{H}_2 = J_2 \sum_j \cos(qa) (\hat{a}_q^\dagger \hat{a}_{-q}^\dagger + \hat{a}_q \hat{a}_{-q}). \tag{2.7.8}$$

The full Hamiltonian is

$$\hat{H} = \sum_q \left[\underbrace{2J_1 \cos(qa)}_{\omega_q} \hat{a}_q^\dagger \hat{a}_q + \underbrace{J_2 \cos(qa)}_{\frac{1}{2}\Delta_q} (\hat{a}_q^\dagger \hat{a}_{-q}^\dagger + \hat{a}_q \hat{a}_{-q}) \right] \tag{2.7.9}$$

Now we carry out canonical transformation as we did in Exercise 2.4:

$$\hat{b}_q = u_q \hat{a}_q + v_q \hat{a}_{-q}^\dagger. \tag{2.7.10}$$

This is slightly different but you can check that it does transform the Hamiltonian in the same way. Note that the q -dependent factor in ω_q and $\frac{1}{2}\Delta_q$ is identical and hence can be factor out, so if the canonical transformation were to diagonalise $\hat{H}$, the u_q and v_q should be independent of q . We will denote them as $u = \cosh \theta$ and $v = \sinh \theta$.

Now the condition for $\hat{H}$ to be diagonalisable is

$$\tanh 2\theta = \frac{\omega_q}{\Delta_q} = \frac{J_2}{J_1} \in (-1, 1). \quad (2.7.11)$$

This gives

$$\hat{H} = \sum_q \tilde{\omega}_q \left(\hat{b}_q \hat{b}_{-q} + \frac{1}{2} \right), \quad (2.7.12)$$

where

$$\tilde{\omega}_q = \sqrt{\omega_q^2 - \Delta_q^2} = 2 \cos(qa) \sqrt{J_2^2 - J_1^2}. \quad (2.7.13)$$

As discussed in Exercise 2.4, when $J_1 = J_2$, the transition is singular and

$$\begin{aligned} \hat{H} &= J \sum_j (\hat{a}_{j+1} + \hat{a}_{j+1}^\dagger)(\hat{a}_j + \hat{a}_j^\dagger) \\ &= 2J \sum_j \hat{x}_{j+1} \hat{x}_j. \end{aligned} \quad (2.7.14)$$

Exercise 2.8

(a) From Schrödinger equation,

$$i\hbar \partial_t e^{-i\hat{H}_0 t/\hbar} |\psi_I(t)\rangle = (\hat{H}_0 + \hat{V}(t)) e^{-i\hat{H}_0 t/\hbar} |\psi_I(t)\rangle. \quad (2.8.1)$$

Expanding the derivative,

$$i\hbar \left[-\frac{i\hat{H}_0}{\hbar} e^{-i\hat{H}_0 t/\hbar} + e^{-i\hat{H}_0 t/\hbar} \partial_t \right] |\psi_I(t)\rangle = (\cancel{\hat{H}_0} + \hat{V}(t)) e^{-i\hat{H}_0 t/\hbar} |\psi_I(t)\rangle, \quad (2.8.2)$$

giving

$$i\hbar e^{-i\hat{H}_0 t/\hbar} \partial_t |\psi_I(t)\rangle = \hat{V}(t) e^{-i\hat{H}_0 t/\hbar} |\psi_I(t)\rangle, \quad (2.8.3)$$

and so

$$i\hbar \partial_t |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle, \quad (2.8.4)$$

where

$$\begin{aligned} \hat{V}_I(t) &= e^{+i\hat{H}_0 t/\hbar} (-f(t)\hat{x}) e^{-i\hat{H}_0 t/\hbar} \\ &= -f(t)\hat{x}(t). \end{aligned} \quad (2.8.5)$$

(b) Solving

$$i\hbar \partial_t |\psi_I(t)\rangle = \hat{V}_I(t) |\psi_I(t)\rangle \quad (2.8.6)$$

iteratively gives

$$\begin{aligned} |\psi_I(t)\rangle &= |\psi_I(-\infty)\rangle + \int_{-\infty}^t dt' \partial_{t'} |\psi_I(t')\rangle \\ &= |0\rangle - \frac{i}{\hbar} \int_{-\infty}^t dt' \hat{V}_I(t') \underbrace{|\psi_I(t')\rangle}_{|0\rangle - \frac{i}{\hbar} \int_{-\infty}^{t'} dt'' \hat{V}_I(t'') |\psi_I(t'')\rangle}. \end{aligned} \quad (2.8.7)$$

Truncating this series to the $O(\hat{V}_I)$ gives

$$\begin{aligned} |\psi_I(t)\rangle &= |0\rangle - \frac{i}{\hbar} \int_{-\infty}^t dt' \hat{V}_I(t') |0\rangle + O(\hat{V}_I^2) \\ &= |0\rangle + \frac{i}{\hbar} \int_{-\infty}^t dt' f(t') \hat{x}(t') |0\rangle + O(f^2), \end{aligned} \quad (2.8.8)$$

and

$$\langle \psi_I(t) | = \langle 0 | - \frac{i}{\hbar} \int_{-\infty}^t dt' f(t') \langle 0 | \hat{x}(t') + O(f^2). \quad (2.8.9)$$

(c)

$$\begin{aligned} \langle \hat{x}(t) \rangle &= \langle \psi_I(t) | \hat{x}(t) | \psi_I(t) \rangle \\ &= \langle 0 | \hat{x}(t) | 0 \rangle - \frac{i}{\hbar} \int_{-\infty}^t dt' f(t') \langle 0 | \hat{x}(t') \hat{x}(t) | 0 \rangle + \frac{i}{\hbar} \int_{-\infty}^t dt'' f(t'') \langle 0 | \hat{x}(t) \hat{x}(t'') | 0 \rangle + O(f^2) \\ &= \underbrace{\langle 0 | \hat{x}(t) | 0 \rangle}_0 + \frac{i}{\hbar} \int_{-\infty}^t dt' f(t') \underbrace{\langle 0 | [\hat{x}(t) \hat{x}(t')] | 0 \rangle}_{R(t-t')} + O(f^2) \\ &= \int_{-\infty}^t dt' R(t-t') f(t') + O(f^2). \end{aligned} \quad (2.8.10)$$

If we define $f(t)$ such that it is switched on only after $t = 0$, and define $R(\tau) = 0$ for $\tau < 0$, then we can rewrite

$$\langle \hat{x}(t) \rangle = \int_0^\infty dt' R(t-t') f(t') + O(f^2). \quad (2.8.11)$$

3 Conserved Particles

Exercise 3.1

(a) For $\hat{c}_3^\dagger \hat{c}_6 \hat{c}_4 \hat{c}_6^\dagger \hat{c}_3 |11111000\dots\rangle$, the first $\hat{c}_3$ removes the particle at state 3, but then $\hat{c}_6^\dagger$ creates a particle at state 6, which is already occupied. This is not allowed for fermions, so

$$\hat{c}_3^\dagger \hat{c}_6 \hat{c}_4 \hat{c}_6^\dagger \hat{c}_3 |11111000\dots\rangle = 0. \quad (3.1.1)$$

(b) To get from $|1111100000\dots\rangle$ to $|1101100100\dots\rangle$, we would like to remove a particle at state 3, then create a particle at state 8, so

$$|1101100100\dots\rangle = (-1)^P \hat{c}_8^\dagger \hat{c}_3 |1111100000\dots\rangle, \quad (3.1.2)$$

with the sign yet to be determined. While acting $\hat{c}_3$ on the Fermi sea state,

$$\hat{c}_3 |1111100000\dots\rangle \equiv \hat{c}_3 \hat{c}_5^\dagger \hat{c}_4^\dagger \hat{c}_3^\dagger \hat{c}_2^\dagger \hat{c}_1^\dagger |0\rangle, \quad (3.1.3)$$

we need to move $\hat{c}_3$ through two creation operators of higher state, each time we get a minus sign. Therefore, the overall parity to destroy the particle at state 3 is positive. Then we can create the particle at state 8, which is the highest state so we don't need to move anything. Therefore,

$$|1101100100\dots\rangle = +\hat{c}_8^\dagger \hat{c}_3 |1111100000\dots\rangle. \quad (3.1.4)$$

(c)

$$\begin{aligned} \langle \Psi | \hat{N} | \Psi \rangle &= |A|^2 \langle 100 | \hat{N} | 100 \rangle + |B|^2 \langle 111000 | \hat{N} | 111000 \rangle \\ &\quad + A^* B \langle 100 | \hat{N} | 111000 \rangle + B^* A \langle 111000 | \hat{N} | 100 \rangle \\ &= |A|^2 + 3 |B|^2. \end{aligned} \quad (3.1.5)$$

If the state is normalised (as it should to make this problem meaningful), $|A|^2 + |B|^2 = 1$, so

$$\langle \Psi | \hat{N} | \Psi \rangle = 1 + 2 |B|^2. \quad (3.1.6)$$

Exercise 3.2

This is similar to Exercise 2.4, just now we have fermions instead of bosons.

(a) We have a lot of canonical anticommutation relations to check between 4 operators! But luckily, most of them vanish (as they should) by the anticommutation relations of a . The only possibly non-vanishing ones are those pairing $\hat{a}_i$ with $\hat{a}_i^\dagger$: they are

$$\{\hat{c}_1, \hat{c}_2\}, \{\hat{c}_1, \hat{c}_1^\dagger\}, \{\hat{c}_2, \hat{c}_2^\dagger\}, \{\hat{c}_1^\dagger, \hat{c}_2^\dagger\}, \quad (3.2.1)$$

in which we want $\{\hat{c}_1, \hat{c}_1^\dagger\} = \{\hat{c}_2, \hat{c}_2^\dagger\} = 1$ and the other two to be zero.

We have

$$\begin{aligned} \{\hat{c}_1, \hat{c}_2\} &= -uv \underbrace{\{\hat{a}_1, \hat{a}_1^\dagger\}}_1 + uv \underbrace{\{\hat{a}_2, \hat{a}_2^\dagger\}}_2 \\ &= 0 \end{aligned} \quad (3.2.2)$$

and

$$\{\hat{c}_1^\dagger, \hat{c}_2^\dagger\} = (\{\hat{c}_1, \hat{c}_2\})^\dagger = 0 \quad (3.2.3)$$

as expected.

$$\begin{aligned} \{\hat{c}_1, \hat{c}_1^\dagger\} &= u^2 \{\hat{a}_1, \hat{a}_1^\dagger\} + v^2 \{\hat{a}_2^\dagger, \hat{a}_2\} \\ &= u^2 + v^2, \end{aligned} \quad (3.2.4)$$

$$\begin{aligned} \{\hat{c}_2, \hat{c}_2^\dagger\} &= v^2 \{\hat{a}_1^\dagger, \hat{a}_1\} + u^2 \{\hat{a}_2, \hat{a}_2^\dagger\} \\ &= u^2 + v^2. \end{aligned} \quad (3.2.5)$$

Therefore the anticommutator relations of c is satisfied if and only if $u^2 + v^2 = 1$.

(b) The Hamiltonian is

$$\hat{H} = \epsilon(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2 \hat{a}_2^\dagger) + \Delta(\hat{a}_1^\dagger \hat{a}_2^\dagger + \text{H.C.}). \quad (3.2.6)$$

The first term is

$$\begin{aligned} \hat{a}_1^\dagger \hat{a}_1 &= (u\hat{c}_1^\dagger - v\hat{c}_2)(u\hat{c}_1 - v\hat{c}_2^\dagger) \\ &= u^2 \hat{c}_1^\dagger \hat{c}_1 - uv \hat{c}_1^\dagger \hat{c}_2^\dagger - vu \hat{c}_2 \hat{c}_1 + v^2 \underbrace{\hat{c}_2 \hat{c}_2^\dagger}_{1 - \hat{c}_2^\dagger \hat{c}_2}, \end{aligned} \quad (3.2.7)$$

and

$$\begin{aligned} \hat{a}_2 \hat{a}_2^\dagger &= (v\hat{c}_1^\dagger + u\hat{c}_2)(v\hat{c}_1 + u\hat{c}_2^\dagger) \\ &= v^2 \hat{c}_1^\dagger \hat{c}_1 + uv \hat{c}_1^\dagger \hat{c}_2^\dagger + uv \hat{c}_2 \hat{c}_1 + u^2 \hat{c}_2 \hat{c}_2^\dagger, \end{aligned} \quad (3.2.8)$$

and so

$$\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2 \hat{a}_2^\dagger = (u^2 - v^2)(\hat{c}_1^\dagger \hat{c}_1 + \hat{c}_2^\dagger \hat{c}_2 - 1) - 2uv(\hat{c}_1^\dagger \hat{c}_2^\dagger + \hat{c}_2 \hat{c}_1). \quad (3.2.9)$$

The second term is

$$\begin{aligned} \hat{a}_1^\dagger \hat{a}_2^\dagger &= (u\hat{c}_1^\dagger - v\hat{c}_2)(v\hat{c}_1 + u\hat{c}_2^\dagger) \\ &= uv \hat{c}_1^\dagger \hat{c}_1 + u^2 \hat{c}_1^\dagger \hat{c}_2^\dagger - v^2 \hat{c}_2 \hat{c}_1 - \underbrace{uv \hat{c}_2 \hat{c}_2^\dagger}_{+uv \hat{c}_2^\dagger \hat{c}_2 - uv}, \end{aligned} \quad (3.2.10)$$

and its Hermitian conjugate is

$$\hat{a}_2 \hat{a}_1 = uv \hat{c}_1^\dagger \hat{c}_1 + u^2 \hat{c}_2 \hat{c}_1 - v^2 \hat{c}_1^\dagger \hat{c}_2^\dagger - vu \hat{c}_2 \hat{c}_2^\dagger, \quad (3.2.11)$$

so

$$\hat{a}_1^\dagger \hat{a}_2^\dagger + \text{H.C.} = 2uv(\hat{c}_1^\dagger \hat{c}_1 + \hat{c}_2^\dagger \hat{c}_2 - 1) + (u^2 - v^2)(\hat{c}_1^\dagger \hat{c}_2^\dagger + \hat{c}_2 \hat{c}_1). \quad (3.2.12)$$

Therefore, the Hamiltonian is

$$\hat{H} = [\epsilon(u^2 - v^2) + 2\Delta uv](\hat{c}_1^\dagger \hat{c}_1 + \hat{c}_2^\dagger \hat{c}_2 - 1) + [\Delta(u^2 - v^2) - 2\epsilon uv](\hat{c}_1^\dagger \hat{c}_2^\dagger + \hat{c}_2 \hat{c}_1). \quad (3.2.13)$$

The factor for the latter term should vanish for $\hat{H}$ to be diagonalisable. Since $u^2 + v^2 = 1$, we set $u = \cos \theta$ and $v = \sin \theta$, so

$$\Delta \cos 2\theta = \epsilon \sin 2\theta, \quad (3.2.14)$$

and so

$$\frac{\Delta}{\epsilon} = \tan 2\theta \in \mathbb{R}. \quad (3.2.15)$$

Therefore, the Hamiltonian for $\epsilon \neq 0$.

We have

$$\sin 2\theta = \frac{\Delta}{\sqrt{\Delta^2 + \epsilon^2}} \quad \cos 2\theta = \frac{\epsilon}{\sqrt{\Delta^2 + \epsilon^2}} \quad (3.2.16)$$

and so

$$\epsilon(u^2 - v^2) + 2\Delta uv = \sqrt{\Delta^2 + \epsilon^2}. \quad (3.2.17)$$

The Hamiltonian is

$$\hat{H} = \sqrt{\Delta^2 + \epsilon^2}(\hat{c}_1^\dagger \hat{c}_1 + \hat{c}_2^\dagger \hat{c}_2 - 1). \quad (3.2.18)$$

(c)

$$\hat{H} |\tilde{0}\tilde{0}\rangle = -\sqrt{\Delta^2 + \epsilon^2} |\tilde{0}\tilde{0}\rangle. \quad (3.2.19)$$

(d) I believe that there is a typo in this question. It should be expanding the c -vacuum state $|\tilde{0}\tilde{0}\rangle$ in terms of the a -fermion states $|n_1 n_2\rangle$:

$$|\tilde{0}\tilde{0}\rangle = b_1 |00\rangle + b_2 |10\rangle + b_3 |01\rangle + b_4 |11\rangle. \quad (3.2.20)$$

We require that $\hat{c}_i |\tilde{0}\tilde{0}\rangle = 0$ for $i = 1, 2$.

$$\begin{aligned} \hat{c}_1 |\tilde{0}\tilde{0}\rangle &= (u\hat{a}_1 + v\hat{a}_2) |\tilde{0}\tilde{0}\rangle \\ &= ub_2 |00\rangle + (vb_1 - ub_4) |01\rangle + vb_2 |11\rangle, \end{aligned} \quad (3.2.21)$$

which requires

$$\begin{cases} b_2 = 0 \\ ub_4 - vb_1 = 0. \end{cases} \quad (3.2.22)$$

$$\begin{aligned} \hat{c}_2 |\tilde{0}\tilde{0}\rangle &= (-v\hat{a}_1 + u\hat{a}_2) |\tilde{0}\tilde{0}\rangle \\ &= ub_3 |00\rangle - (vb_1 - ub_4) |01\rangle + vb_3 |11\rangle, \end{aligned} \quad (3.2.23)$$

which requires

$$\begin{cases} b_3 = 0 \\ ub_4 - vb_1 = 0. \end{cases} \quad (3.2.24)$$

This is solved by $b_1 = u$ and $b_4 = v$. Therefore

$$\begin{aligned} |\tilde{0}\tilde{0}\rangle &= u |00\rangle + v |11\rangle \\ &= (u + v\hat{a}_2^\dagger \hat{a}_1^\dagger) |00\rangle. \end{aligned} \quad (3.2.25)$$

Exercise 3.3

(a) Substituting

$$\rho(\mathbf{x}) = \sum_{j=1}^N \delta(\mathbf{x} - \mathbf{x}_j), \quad (3.3.1)$$

the right hand side can be expanded as

$$\begin{aligned}
\text{RHS} &= \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) \left[\sum_i \delta(\mathbf{x} - \mathbf{x}_i) \right] \left[\sum_j \delta(\mathbf{y} - \mathbf{x}_j) \right] - \frac{1}{2} V(0)N \\
&= \sum_{j < i} \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) \delta(\mathbf{x} - \mathbf{x}_i) \delta(\mathbf{y} - \mathbf{x}_j) \\
&\quad + \frac{1}{2} \sum_{i=1}^N \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) \delta(\mathbf{x} - \mathbf{x}_i) \delta(\mathbf{y} - \mathbf{x}_i) - \frac{1}{2} V(0)N \\
&= \sum_{i < j} V(\mathbf{x}_i - \mathbf{x}_j) + \frac{1}{2} V(0)N - \frac{1}{2} V_0 N \\
&= \sum_{i < j} V(\mathbf{x}_i - \mathbf{x}_j) = V.
\end{aligned} \tag{3.3.2}$$

We see that $\frac{1}{2} V_0 N$ is exactly the self-interaction energy.

(b) Direct second quantisation gives

$$\begin{aligned}
\hat{V} &= \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) \hat{\rho}(\mathbf{x}) \hat{\rho}(\mathbf{y}) - \frac{1}{2} \int d^3\mathbf{x} V(0) \hat{\rho}(\mathbf{x}) \\
&= \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) \hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}(\mathbf{x}) \hat{\phi}^\dagger(\mathbf{y}) \hat{\phi}(\mathbf{y}) - \frac{1}{2} \int d^3\mathbf{x} V(0) \hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}(\mathbf{x}).
\end{aligned} \tag{3.3.3}$$

We have

$$\begin{aligned}
\hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}(\mathbf{x}) \hat{\phi}^\dagger(\mathbf{y}) \hat{\phi}(\mathbf{y}) &= \hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}(\mathbf{y}) \delta(\mathbf{x} - \mathbf{y}) \pm \hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}^\dagger(\mathbf{y}) \hat{\phi}(\mathbf{x}) \hat{\phi}(\mathbf{y}) \\
&= \hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}(\mathbf{y}) \delta(\mathbf{x} - \mathbf{y}) + \underbrace{\hat{\phi}^\dagger(\mathbf{y}) \hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}(\mathbf{x}) \hat{\phi}(\mathbf{y})}_{:\hat{\rho}(\mathbf{x})\hat{\rho}(\mathbf{y}):}.
\end{aligned} \tag{3.3.4}$$

Therefore,

$$\begin{aligned}
\hat{V} &= \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) : \hat{\rho}(\mathbf{x}) \hat{\rho}(\mathbf{y}) : \\
&\quad + \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) \hat{\phi}^\dagger(\mathbf{x}) \hat{\phi}(\mathbf{y}) \delta(\mathbf{x} - \mathbf{y}) - \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(0) \hat{\rho}(\mathbf{x}) \\
&= \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(\mathbf{x} - \mathbf{y}) : \hat{\rho}(\mathbf{x}) \hat{\rho}(\mathbf{y}) :,
\end{aligned} \tag{3.3.5}$$

where the last two terms cancel.

Exercise 3.4

(a)

$$\hat{V} = \frac{1}{2} \int d^3\mathbf{x} d^3\mathbf{y} V(\|\mathbf{x} - \mathbf{y}\|) : \hat{\rho}(\mathbf{x}) \hat{\rho}(\mathbf{y}) :. \tag{3.4.1}$$

(b) Commutation relation in momentum basis:

$$\begin{aligned}
[\hat{c}_{\mathbf{k}}, \hat{c}_{\mathbf{k}'}^\dagger] &= \int d^3\mathbf{r} d^3\mathbf{r}' \underbrace{[\hat{\phi}(\mathbf{r}), \hat{\phi}^\dagger(\mathbf{r}')]_{\pm}}_{\delta(\mathbf{r} - \mathbf{r}')} e^{-i\mathbf{k} \cdot \mathbf{r}} e^{i\mathbf{k}' \cdot \mathbf{r}'} \\
&= \int d^3\mathbf{r} e^{-i\mathbf{k} \cdot \mathbf{r}} e^{i\mathbf{k}' \cdot \mathbf{r}} \\
&= (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}').
\end{aligned} \tag{3.4.2}$$

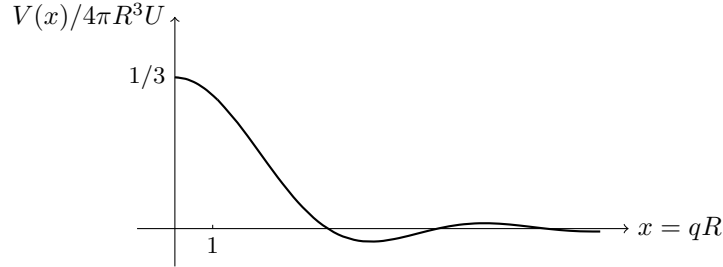
Fourier transform of potential:

$$\begin{aligned}
 V(\mathbf{q}) &= \int d^3\mathbf{r} V(\mathbf{r}) e^{-i\mathbf{q}\cdot\mathbf{r}} \\
 &= \int r^2 \sin\theta dr d\theta d\phi V(r) e^{-iqr \cos\theta} \\
 &= 2\pi \int_0^\infty dr r^2 V(r) \int_{-1}^1 d(\cos\theta) e^{-iqr \cos\theta} \\
 &= \frac{4\pi U}{q} \int_0^R dr r \sin qr \\
 &= \frac{4\pi U}{q^3} [\sin qR - qR \cos qR] .
 \end{aligned} \tag{3.4.3}$$

The dimensionless form is

$$\frac{V(x)}{4\pi R^3 U} = \frac{\sin x - x \cos x}{x^3}, \tag{3.4.4}$$

where $x = qR$.



Exercise 3.5

(a) Expanding the trace in the N -particle energy eigenstate,

$$\begin{aligned}
 F &= -k_B T \ln \text{tr}[e^{-\beta(\hat{H} - \mu \hat{N})}] \\
 &= -k_B T \ln \left(\sum_n e^{-\beta(E_n - \mu N)} \right) .
 \end{aligned} \tag{3.5.1}$$

The partial derivative with respect to μ is

$$\begin{aligned}
 -\frac{\partial F}{\partial \mu} &= k_B T \cdot \frac{\sum_n \frac{\partial}{\partial \mu} e^{-\beta(E_n - \mu N)}}{\sum_n e^{-\beta(E_n - \mu N)}} \\
 &= k_B T \cdot \frac{\sum_n \beta N e^{-\beta(E_n - \mu N)}}{\sum_n e^{-\beta(E_n - \mu N)}} \\
 &= N .
 \end{aligned} \tag{3.5.2}$$

(b) Considering the single level as a grand canonical ensemble, the grand partition function is

$$\Xi = \sum_{n=0}^{\infty} e^{-\beta(\epsilon - \mu)n} = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \tag{3.5.3}$$

where we denoted $x = e^{-\beta(\epsilon - \mu)}$. We must have $x < 1$, or $\mu < \epsilon$, for convergence.

The average particle number in this grand ensemble is given by

$$\langle \hat{n} \rangle = -\frac{\partial \Omega}{\partial \mu} = \frac{1}{\beta} \frac{\partial \ln \Xi}{\partial \mu}, \quad (3.5.4)$$

where $\Omega = -k_B T \ln \Xi$ is the grand (Landau) potential.² Therefore,

$$\begin{aligned} \langle \hat{n} \rangle &= \frac{1}{\beta \Xi} \frac{\partial \Xi}{\partial \mu} \\ &= \frac{1}{e^{\beta(\epsilon - \mu)} - 1}. \end{aligned} \quad (3.5.5)$$

Exercise 3.6

(a) Writing the delta functions in Laplace transforms, the microcanonical density of state is

$$\begin{aligned} W(E, N) &= \text{tr}[\delta(E - \hat{H})\delta(N - \hat{N})] \\ &= \int_{\beta_0 - i\infty}^{\beta_0 + i\infty} \frac{d\beta}{2\pi i} \int_{\lambda_0 - i\infty}^{\lambda_0 + i\infty} \frac{d\lambda}{2\pi i} e^{+\beta E} e^{-\lambda N} \text{tr}[e^{-\beta \hat{H}} e^{+\lambda \hat{N}}]. \end{aligned} \quad (3.6.1)$$

If we identify $\lambda = \beta\mu$, then

$$\text{tr}[e^{-\beta(\hat{H} - \mu \hat{N})}] = \Xi(\beta, \mu) \quad (3.6.2)$$

is the grand partition function, and

$$\begin{aligned} W(E, N) &= \int \frac{d\beta}{2\pi i} \int \frac{d\lambda}{2\pi i} e^{\beta(E - \mu N)} \Xi \\ &= \int \frac{d\beta}{2\pi i} \int \frac{d\lambda}{2\pi i} e^{\beta(E - \mu N) + \ln \Xi} \\ &=: \int \frac{d\beta}{2\pi i} \int \frac{d\lambda}{2\pi i} e^{\Phi(\beta, \mu)}. \end{aligned} \quad (3.6.3)$$

All terms in $\Phi = \beta(E - \mu N) + \ln \Xi$ are extensive, so in a macroscopic system, this integral is dominated by its very sharp peak at $\Phi(\beta^*, \mu^*)$, where

$$\left. \frac{\partial \Phi}{\partial \beta} \right|_{(\beta^*, \mu^*)} = \left. \frac{\partial \Phi}{\partial \lambda} \right|_{(\beta^*, \mu^*)} = 0, \quad (3.6.4)$$

i.e. when

$$E = -\frac{\partial \ln \Xi}{\partial \beta} = \langle \hat{H} \rangle, \quad (3.6.5)$$

$$N = -\frac{\partial \ln \Xi}{\partial (\beta\mu)} = \langle \hat{N} \rangle. \quad (3.6.6)$$

This is to say that in the thermodynamic limit, a microcanonical and a grand canonical ensemble is equivalent, with (E, N) in the microcanonical ensemble the same as $(\langle \hat{H} \rangle, \langle \hat{N} \rangle)$ in the grand canonical ensemble.

Therefore,

$$\begin{aligned} \ln W &= \Phi(\beta^*, \lambda^*) + O(\ln N) \\ &= \ln \Xi - \beta^*(E - \mu^* N) + O(\ln N). \end{aligned} \quad (3.6.7)$$

²The proof is similar to the above one in canonical ensemble with conserved particle, just now we also need to sum over the particle numbers in expanding the trace.

For thermodynamics, we have

$$\Omega = -k_B T \ln \Xi = -k_B T \ln \Xi = E - TS - \mu N. \quad (3.6.8)$$

Therefore,

$$\begin{aligned} S &= k_B (\ln \Xi - \beta E - \beta \mu N) \\ &= k_B \ln W. \end{aligned} \quad (3.6.9)$$

(b) The microcanonical average is

$$\begin{aligned} \langle A \rangle_M &= \frac{\text{tr}[\delta(E - \hat{H})\delta(N - \hat{N})\hat{A}]}{W(E, N)} \\ &= \frac{\int \frac{d\beta d\lambda}{(2\pi i)} e^{\beta E - \lambda N} \text{tr}[e^{-\beta \hat{H} + \lambda \hat{N}} \hat{A}]}{\int \frac{d\beta d\lambda}{(2\pi i)} e^{\beta E - \lambda N} \text{tr}[e^{-\beta \hat{H} + \lambda \hat{N}}]} \\ &= \frac{\int \frac{d\beta d\lambda}{(2\pi i)} e^{\beta E - \lambda N} \Xi(\beta, \mu) \langle \hat{A} \rangle_{\beta, \lambda}}{\int \frac{d\beta d\lambda}{(2\pi i)} e^{\beta E - \lambda N} \Xi(\beta, \mu)} \\ &= \frac{\Xi(\beta^*, \mu^*) \langle \hat{A} \rangle_{\beta^*, \lambda^*}}{\Xi(\beta^*, \mu^*)} \\ &= \langle \hat{A} \rangle_{\beta^*, \mu^*}, \end{aligned} \quad (3.6.10)$$

which is the grand canonical average at equilibrium, where we again assumed $e^{\Phi(\beta, \lambda)}$ to be sharply peaked in the thermodynamic limit.

Also,

$$\begin{aligned} \langle A \rangle_M &= \frac{\Xi(\beta^*, \mu^*) \langle \hat{A} \rangle_{\beta^*, \lambda^*}}{\Xi(\beta^*, \mu^*)} \\ &= \frac{1}{\Xi(\beta^*, \mu^*)} \text{tr}[e^{-\beta^* (\hat{H} - \mu^* \hat{N})} \hat{A}] \\ &= \text{tr}[\hat{\rho}_B \hat{A}]. \end{aligned} \quad (3.6.11)$$

4 Simple Examples of Second Quantization

Exercise 4.1

(a) Expanding the Hamiltonian gives³

$$\begin{aligned}\hat{H} &= - \sum_j \left[\frac{J_1}{4} (S_j^+ + S_j^-) (S_{j+1}^+ S_{j+1}^-) - \frac{J_2}{4} (S_j^+ - S_j^-) (S_{j+1}^+ - S_{j+1}^-) \right] \\ &= - \sum_j \left[t (S_j^+ + S_{j+1}^- + S_j^- S_{j+1}^+) - \Delta (S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+) \right].\end{aligned}\quad (4.1.1)$$

Now we perform the Jordan–Wigner transformation. Pay attention to the signs while commuting the operators. We have

$$\begin{aligned}S_j^+ S_{j+1}^+ &= d_j^\dagger e^{i\phi_j} d_{j+1}^\dagger e^{i\phi_{j+1}} \\ &= d_j^\dagger e^{i\pi n_j} d_{j+1}^\dagger \\ &= d_j^\dagger d_{j+1}^\dagger = -d_{j+1}^\dagger d_j^\dagger.\end{aligned}\quad (4.1.2)$$

$$\begin{aligned}S_j^- S_{j+1}^- &= d_j e^{i\phi_j} d_{j+1} e^{i\phi_{j+1}} \\ &= d_j e^{i\pi n_j} d_{j+1} \\ &= -d_j d_{j+1}.\end{aligned}\quad (4.1.3)$$

$$\begin{aligned}S_j^+ S_{j+1}^- &= d_j^\dagger e^{i\phi_j} d_{j+1} e^{i\phi_{j+1}} \\ &= d_j^\dagger e^{i\pi n_j} d_{j+1} \\ &= d_j^\dagger d_{j+1}.\end{aligned}\quad (4.1.4)$$

$$\begin{aligned}S_j^- S_{j+1}^+ &= d_j e^{i\phi_j} d_{j+1}^\dagger e^{i\phi_{j+1}} \\ &= d_j e^{i\pi n_j} d_{j+1}^\dagger \\ &= -d_j d_{j+1}^\dagger = d_{j+1}^\dagger d_j.\end{aligned}\quad (4.1.5)$$

Therefore,

$$\hat{H} = - \sum_j \left[t (d_{j+1}^\dagger d_j + d_j^\dagger d_{j+1}) + \Delta (d_{j+1}^\dagger d_j^\dagger + d_j d_{j+1}) \right]. \quad (4.1.6)$$

(b) Transform into the Fourier basis,

$$d_j = \frac{1}{\sqrt{N}} \sum_k e^{ikja} d_k. \quad (4.1.7)$$

We have

$$\sum_j d_{j+1}^\dagger d_j = \sum_k e^{-ika} d_j^\dagger d_j, \quad (4.1.8)$$

and its Hermitian conjugate is

$$\sum_k e^{ika} d_j^\dagger d_j. \quad (4.1.9)$$

³From now on we will drop the hats on the operators when there is no confusion, as there will rarely be anything classical.

Therefore,

$$\sum_j (d_{j+1}^\dagger d_j + d_j^\dagger d_{j+1}) = \sum_k 2 \cos(ka) d_k^\dagger d_k. \quad (4.1.10)$$

Next,

$$\begin{aligned} \sum_j d_{j+1}^\dagger d_j &= \sum_k e^{-ika} d_k^\dagger d_{-k}^\dagger \\ &= \frac{1}{2} \sum_k (-e^{ika} d_k^\dagger d_{-k}^\dagger + e^{-ika} d_k^\dagger d_{-k}^\dagger) \\ &= -i \sum_k \sin(ka) d_k^\dagger d_{-k}^\dagger, \end{aligned} \quad (4.1.11)$$

and its Hermitian conjugate is

$$-i \sum_k \sin(ka) d_k d_{-k}, \quad (4.1.12)$$

so,

$$\sum_j (d_{j+1}^\dagger d_j^\dagger + d_j d_{j+1}) = -i \sum_k \sin(ka) (d_k^\dagger d_{-k}^\dagger + d_k d_{-k}). \quad (4.1.13)$$

Therefore,

$$\begin{aligned} \hat{H} &= \sum_k \left[-2t \cos(ka) d_k^\dagger d_k + i\Delta (d_k^\dagger d_{-k}^\dagger + d_k d_{-k}) \right] \\ &= -2td_0^\dagger d_0 + \underbrace{\sum_{k>0} \left[\epsilon_k (d_k^\dagger d_k + d_{-k}^\dagger d_{-k}) + i\delta_k (d_k^\dagger d_{-k}^\dagger + d_k d_{-k}) \right]}_{\hat{H}_k}, \end{aligned} \quad (4.1.14)$$

where

$$\begin{cases} \epsilon_k = -2t \cos(ka) \\ \delta_k = 2\Delta \sin(ka) \end{cases} \quad (4.1.15)$$

Each $\hat{H}_k$ in the sum is similar to the Hamiltonian we met in Exercise 3.2, except now there is a factor of i in front of the pairing term. To account for this, we modify the Bogoliubov transform to

$$\begin{cases} \gamma_k = u_k d_k + i v_k d_{-k}^\dagger \\ \gamma_{-k} = u_k d_{-k} - i v_k d_k^\dagger \end{cases} \quad (4.1.16)$$

with u_k, v_k real and $u_k^2 + v_k^2 = 1$, so that we can write $u_k = \cos \theta_k$, $v_k = \sin \theta_k$. Repeating the process of Exercise 3.2, we get

$$\tan 2\theta_k = \frac{\delta_k}{\epsilon_k} = -\frac{\Delta}{t} \tan(ka) \in \mathbb{R}, \quad (4.1.17)$$

so that

$$\begin{cases} \sin 2\theta_k = \frac{\delta_k}{\sqrt{\epsilon_k^2 + \delta_k^2}} \\ \cos 2\theta_k = \frac{\epsilon_k}{\sqrt{\epsilon_k^2 + \delta_k^2}} \end{cases} \quad (4.1.18)$$

The diagonalised Hamiltonian is

$$\hat{H}_k = \epsilon_k (\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1), \quad (4.1.19)$$

where the energy of each mode is

$$\begin{aligned} \epsilon_k &= \sqrt{\epsilon^2 + \Delta^2} \\ &= 2\sqrt{t^2 \cos^2(ka) + \Delta^2 \sin^2(ka)} \\ &= \frac{1}{2} \sqrt{J_1^2 + J_2^2 + 2J_1 J_2 \cos(2ka)}. \end{aligned} \quad (4.1.20)$$

Note that is energy ϵ_k is non-negative, and should be interpreted as the excitation energy. Defining the excitation operator of the zero mode by

$$d_0^\dagger d_0 = 1 - \gamma_0^\dagger \gamma_0, \quad (4.1.21)$$

the diagonalised Hamiltonian is

$$\begin{aligned} \hat{H} &= -t + \sum_k \epsilon_k \left(\gamma_k^\dagger \gamma_k - \frac{1}{2} \right) \\ &= E_{\text{GS}} + \sum_k \epsilon_k \gamma_k^\dagger \gamma_k, \end{aligned} \quad (4.1.22)$$

where E_{GS} is the ground state energy, and ϵ_k is the excitation spectrum.

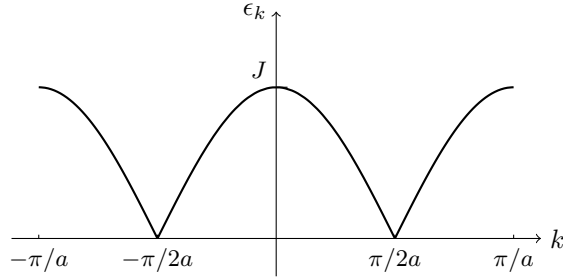
1. $J_1 = J_2 = J$. Then

$$\hat{H} = -J \sum_k \cos(ka) d_k^\dagger d_k. \quad (4.1.23)$$

The excitation spectrum is

$$\epsilon_k = |\cos(ka)|, \quad (4.1.24)$$

as shown in the main text. This spectrum is gapless at $k = \pm \frac{\pi}{2a}$.



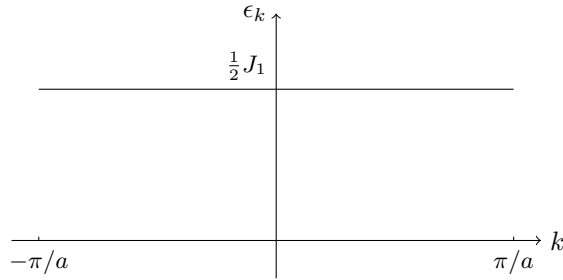
2. $J_2 = 0$. Then

$$\hat{H} = -J_1 \sum_j S_j^x S_{j+1}^x \quad (4.1.25)$$

is just the Ising Hamiltonian. The excitation spectrum is

$$\epsilon_k = \frac{1}{2} |J_1|, \quad (4.1.26)$$

which is constant.

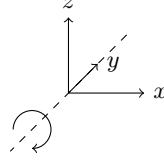


Exercise 4.2

(a) Rotate 90° about y axis transforms $S_x \rightarrow S_z$ and $S_z \rightarrow -S_x$, and hence

$$\hat{H} = -J \sum_j S_x(j) S_x(j+1) - h \sum_j S_z(j). \quad (4.2.1)$$

More formally, the $J = \frac{1}{2}$ representation of rotation about y axis is



$$U_y(\theta) = e^{-i\theta S_y} = e^{-i\theta \sigma_y/2}, \quad (4.2.2)$$

and so the spin operators transform as

$$\begin{aligned} U_y(\theta) S_x U_y^\dagger &= S_x \cos \theta - S_z \sin \theta \\ U_y(\theta) S_z U_y^\dagger &= S_z \cos \theta + S_x \sin \theta. \end{aligned} \quad (4.2.3)$$

(b) Have

$$\begin{aligned} S_x(j) S_x(j+1) &= \frac{1}{4} (S_j^+ + S_j^-) (S_{j+1}^+ + S_{j+1}^-) \\ &= \frac{1}{4} (S_j^+ S_{j+1}^+ + S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+ + S_j^- S_{j+1}^-). \end{aligned} \quad (4.2.4)$$

Jordan–Wigner transform gives

$$\begin{aligned} S_x(j) S_x(j+1) &= \frac{1}{4} (f_j^+ f_{j+1}^+ + f_j f_{j+1} = f_j f_{j+1}^+ - f_j f_{j+1}) \\ &= -\frac{1}{4} (f_j - f_j^+) (f_{j+1} + f_{j+1}^+) \end{aligned} \quad (4.2.5)$$

The Jordan–Wigner transform of $S_z(j)$ is

$$S_z(j) = f_j^\dagger f_j - \frac{1}{2}. \quad (4.2.6)$$

Therefore, the transformed Hamiltonian is

$$\hat{H} = \frac{J}{4} \sum_j (f_j - f_j^+) (f_{j+1} + f_{j+1}^+) - h \sum_j f_j^\dagger f_j, \quad (4.2.7)$$

where the constant term $+Nh/2$ is dropped.

(c) Transforming to the Fourier basis via

$$f_j = \frac{1}{\sqrt{N}} \sum_k d_k e^{ikR_j} \quad (4.2.8)$$

gives

$$\begin{aligned} \sum_j (f_j - f_j^+)(f_{j+1} + f_{j+1}^+) &= \sum_k (e^{-ika} d_k d_{-k} + e^{-ika} d_k d_k^\dagger - e^{ika} d_k^\dagger d_k - e^{ika} d_k^\dagger d_{-k}^\dagger) \\ &= \sum_k -i \sin(ka) d_k d_{-k} - 2 \cos(ka) d_k^\dagger d_k - i \sin(ka) d_k^\dagger d_{-k}^\dagger, \end{aligned} \quad (4.2.9)$$

while

$$\sum_j f_j^\dagger f_j = \sum_k d_k d_{-k} \quad (4.2.10)$$

This gives

$$\begin{aligned} \hat{H} &= \sum_k \left[\epsilon_k d_k^\dagger d_k - \frac{1}{2} i \Delta_k (d_k^\dagger d_{-k}^\dagger + d_k d_{-k}) \right] \\ &= \sum_k \left[\epsilon_k (d_k^\dagger d_k - d_{-k} d_{-k}^\dagger) - i \Delta_k (d_k^\dagger d_{-k}^\dagger + d_k d_{-k}) \right], \end{aligned} \quad (4.2.11)$$

where

$$\begin{cases} \epsilon_k = -\frac{J}{2} \cos(ka) - h \\ \Delta_k = \frac{J}{2} \sin(ka). \end{cases} \quad (4.2.12)$$

(d) Using the same transformation as in Exercise 4.1 gives

$$\hat{H} = E_{\text{GS}} + \sum_k E_k \gamma_k^\dagger \gamma_k, \quad (4.2.13)$$

where the excitation energy is

$$\begin{aligned} E_k &= \sqrt{\epsilon_k^2 + \Delta_k^2} \\ &= \sqrt{\frac{J^2}{4} + Jh \cos(ka) + h^2} \\ &= \sqrt{\left(h - \frac{J}{2}\right)^2 + 2Jh \cos^2\left(\frac{ka}{2}\right)}. \end{aligned} \quad (4.2.14)$$

This is non-negative, and equals to zero if and only if $h = \frac{J}{2}$ and $k = \frac{\pi}{a}$.

At small h ($h < \frac{J}{2}$), the Ising interaction dominates, while at large h ($h > \frac{J}{2}$), the alignment with the external transverse field dominates. $h = \frac{J}{2}$ is the quantum phase transition point.

Exercise 4.3

We perform the Fourier transform

$$c_{j,\sigma} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}_j} c_{\mathbf{k},\sigma}. \quad (4.3.1)$$

The number operator becomes

$$\sum_j c_{j,\sigma}^\dagger c_{j,\sigma} = \sum_{\mathbf{k}} c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma}. \quad (4.3.2)$$

The hopping term is

$$\sum_{j,\mathbf{a}} c_{j+\mathbf{a},\sigma}^\dagger c_{j,\sigma} = \sum_{\mathbf{k}} (e^{-i\mathbf{k} \cdot \hat{\mathbf{x}}a} + e^{-i\mathbf{k} \cdot \hat{\mathbf{y}}a}) c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma}, \quad (4.3.3)$$

and its Hermitian conjugate is

$$\sum_{\mathbf{k}} (e^{+i\mathbf{k}\cdot\hat{\mathbf{x}}a} + e^{+i\mathbf{k}\cdot\hat{\mathbf{y}}a}) c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma} . \quad (4.3.4)$$

Therefore, the Hamiltonian is

$$\hat{H} = \sum_{\mathbf{k},\sigma} \epsilon(\mathbf{k}) c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma} , \quad (4.3.5)$$

where the dispersion is

$$\epsilon(\mathbf{k}) = -2t(\cos k_x a + \cos k_y a) - \mu . \quad (4.3.6)$$

(b) When $n_e = 2$, the electron fills the whole (first) Brillouin zone, so

$$\frac{n_e}{2} = \frac{A_{\text{FS}}}{A_{\text{BZ}}} . \quad (4.3.7)$$

Therefore, the area of the Fermi surface is

$$A_{\text{FS}} = \frac{n_e}{2} A_{\text{BZ}} = \frac{n_e}{2} \left(\frac{2\pi}{a} \right)^2 = \frac{2\pi^2}{a^2} n_e . \quad (4.3.8)$$

(c) The edge of the Fermi surface satisfies

$$\cos(k_x a) + \cos(k_y a) = -\frac{E_F + \mu}{2t} . \quad (4.3.9)$$

For small n_e (and hence small $\|\mathbf{k}\|$), expanding the cosine gives

$$1 - \frac{1}{2}(k_x a)^2 + 1 - \frac{1}{2}(k_y a)^2 + O(k^4) = -\frac{E_F + \mu}{2t} , \quad (4.3.10)$$

and hence

$$k_x^2 + k_y^2 + O(k^4) = \frac{1}{a^2} \left(4 + \frac{E_F + \mu}{t} \right) . \quad (4.3.11)$$

When k is small enough, the Fermi surface is a circle. As n_e increases, its shape deviates from a circle.

At $n_e = 1$, one have the particle-hole symmetry and hence $\mu = 0$. The equation of the Fermi edge becomes

$$\cos(k_x a) + \cos(k_y a) = 2 \cos\left(\frac{a}{2}(k_x + k_y)\right) \cos\left(\frac{a}{2}(k_x - k_y)\right) = 0 , \quad (4.3.12)$$

which is solved by

$$k_x \pm k_y = \pm \frac{\pi}{a} . \quad (4.3.13)$$

The Fermi surface is a square (diamond).

(d) Let $\mathbf{Q} = (\frac{\pi}{a}, \frac{\pi}{a})$. At half-filling, $\mu = 0$, so

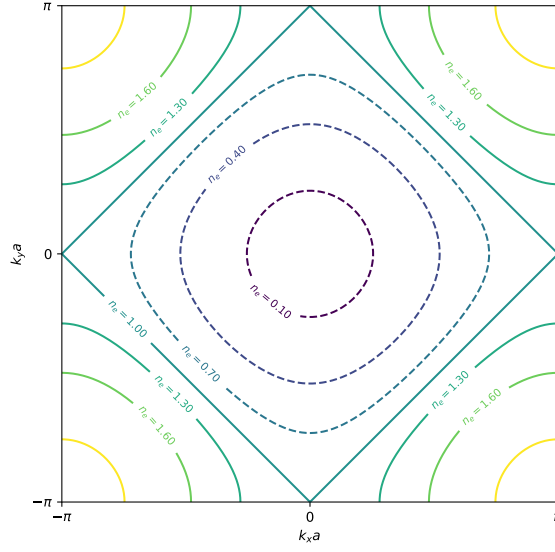
$$\epsilon_{\mathbf{k}} = -2t[\cos(k_x a) + \cos(k_y a)] . \quad (4.3.14)$$

One might note that

$$\epsilon_{\mathbf{k}+\mathbf{Q}} = -\epsilon_{\mathbf{k}} . \quad (4.3.15)$$

Since $+\mathbf{Q}$ transforms the Fermi surface onto itself,

$$\epsilon_{\mathbf{k}+\mathbf{Q}} = \epsilon_{\mathbf{k}} = 0 , \quad (4.3.16)$$

Figure 1: Fermi surface edges with increasing n_e .

and hence there are low-energy excitation

$$c_{\mathbf{k}+\mathbf{Q},\sigma}^\dagger c_{\mathbf{k},\sigma} \quad (4.3.17)$$

with *excitation energy arbitrarily small* near the Fermi surface edge.

Now consider the spin density wave with wavevector $\mathbf{Q}$,

$$\langle S_j^z \rangle = \frac{1}{2} e^{i\mathbf{Q} \cdot \mathbf{R}_j} = \frac{1}{2} (-1)^{j_x + j_y}. \quad (4.3.18)$$

This is just the antiferromagnetic arrangement:

$$\begin{array}{cccc} \downarrow & \uparrow & \downarrow & \uparrow \\ \uparrow & \downarrow & \uparrow & \downarrow \\ \downarrow & \uparrow & \downarrow & \uparrow \\ \uparrow & \downarrow & \uparrow & \downarrow \end{array}$$

In momentum space, this spin wave is

$$\langle S_{\mathbf{Q}}^z \rangle = \frac{1}{2} \sum_{\mathbf{k}} \left(\langle c_{\mathbf{k}+\mathbf{Q},\uparrow}^\dagger c_{\mathbf{k},\uparrow} \rangle - \langle c_{\mathbf{k}+\mathbf{Q},\downarrow}^\dagger c_{\mathbf{k},\downarrow} \rangle \right), \quad (4.3.19)$$

which is exactly the low-energy excitation, and hence the $n_e = 1$ ground state is unstable to the formation of this spin wave.

To see how the role of the interaction

$$\hat{H}_{\text{int}} = U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (4.3.20)$$

we use mean field theory. We define the mean local spin polarisation

$$m_i = \frac{1}{2} (\langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle) \quad (4.3.21)$$

and mean local density

$$n_i = \frac{1}{2}(\langle n_{i\uparrow} \rangle + \langle n_{i\downarrow} \rangle). \quad (4.3.22)$$

The mean-field decoupling approximates

$$\begin{aligned} n_{i\uparrow}n_{i\downarrow} &\approx n_{i\uparrow}\langle n_{i\downarrow} \rangle + \langle n_{i\uparrow} \rangle n_{i\downarrow} - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle \\ &= \frac{n_i}{2}(n_{i\uparrow} + n_{i\downarrow}) - m_i(n_{i\uparrow} - n_{i\downarrow}) + \text{const.}, \end{aligned} \quad (4.3.23)$$

and so

$$\hat{H}_{\text{int}}^{\text{MF}} = U \sum_i \left[\frac{n_i}{2}(n_{i\uparrow} + n_{i\downarrow}) - m_i(n_{i\uparrow} - n_{i\downarrow}) \right]. \quad (4.3.24)$$

Since the interaction penalises double occupation, at half-filling there is roughly one electron per site. The first density term is roughly a constant, and the important part is

$$\hat{H}_{\text{int}}^{\text{MF}} = -U \sum_i m_i(n_{i\uparrow} - n_{i\downarrow}). \quad (4.3.25)$$

Now suppose the spin wave has developed, with

$$m_i = m e^{i\mathbf{Q} \cdot \mathbf{R}_i}, \quad (4.3.26)$$

we have

$$\hat{H}_{\text{SDW}}^{\text{MF}} = U m \sum_i e^{i\mathbf{Q} \cdot \mathbf{R}_i} (n_{i\uparrow} - n_{i\downarrow}). \quad (4.3.27)$$

Taking Fourier transform, this is

$$\hat{H}_{\text{SDW}}^{\text{MF}} = -\Delta \sum_{\mathbf{k}} \left(c_{\mathbf{k}+\mathbf{Q},\uparrow}^\dagger c_{\mathbf{k},\uparrow} - c_{\mathbf{k}+\mathbf{Q},\downarrow}^\dagger c_{\mathbf{k},\downarrow} \right), \quad (4.3.28)$$

coupling $\mathbf{k}$ with $\mathbf{k} + \mathbf{Q}$. $\Delta \sim U$ sets the strength of the self-consistent SDW mixing.

Now we focus on the subspace of an interacting spin-up pair with momenta $\mathbf{k}$ and $\mathbf{k} + \mathbf{Q}$, the Hamiltonian is

$$H_{\mathbf{k}} = \begin{pmatrix} \epsilon_{\mathbf{k}} & \Delta \\ \Delta & -\epsilon_{\mathbf{k}} \end{pmatrix}. \quad (4.3.29)$$

Diagonalising gives the energy after interaction:

$$E_{\pm}(\mathbf{k}) = \pm \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}. \quad (4.3.30)$$

At the Fermi surface with $\epsilon_{\mathbf{k}} = 0$, now

$$E_{\pm}(\mathbf{k}) = \pm |\Delta|, \quad (4.3.31)$$

and hence a band gap of $2|\Delta|$ is opened.

(e) The additional Hamiltonian is

$$\sum_{\sigma, j, \mathbf{a}=(\hat{\mathbf{x}} \pm \hat{\mathbf{y}})a} c_{j+\mathbf{a}, \sigma}^\dagger c_{j, \sigma} = \sum_{\sigma, \mathbf{k}} \left(e^{i(k_x + k_y)a} + e^{i(k_x - k_y)a} \right) c_{\mathbf{k}, \sigma}^\dagger c_{\mathbf{k}, \sigma} \quad (4.3.32)$$

plus its Hermitian conjugate

$$\sum_{\sigma, \mathbf{k}} \left(e^{-i(k_x + k_y)a} + e^{-i(k_x - k_y)a} \right) c_{\mathbf{k}, \sigma}^\dagger c_{\mathbf{k}, \sigma}. \quad (4.3.33)$$

Hence now, the dispersion has an extra term

$$2 [\cos(k_x + k_y)a + \cos(k_x - k_y)a] = 4 \cos(k_x a) \cos(k_y a). \quad (4.3.34)$$

Therefore,

$$\hat{H} = \sum_{\sigma, \mathbf{k}} \epsilon_{\mathbf{k}} \gamma_{\mathbf{k}, \sigma}^{\dagger} \gamma_{\mathbf{k}, \sigma} \quad (4.3.35)$$

with

$$\epsilon_{\mathbf{k}} = -2t[\cos(k_x a) + \cos(k_y a)] - 4t' \cos(k_x a) \cos(k_y a) - \mu. \quad (4.3.36)$$

Now at half-filling, the Fermi surface edge is no longer a square, and $\mathbf{k} + \mathbf{Q}$ no longer results in a perfect nesting. The band gap no longer opens.

Heuristically, these extra new coupling create frustration to the antiferromagnetic spin wave.

Exercise 4.4

(a) Trivial.

(b) Fourier transform gives

$$\sum_j \psi_B^{\dagger}(\mathbf{r}_j) \psi_A(\mathbf{r}_j) = \sum_{\mathbf{k}} c_{\mathbf{k}B}^{\dagger} c_{\mathbf{k}A}, \quad (4.4.1)$$

and its Hermitian conjugate is

$$\sum_{\mathbf{k}} c_{\mathbf{k}A}^{\dagger} c_{\mathbf{k}B}. \quad (4.4.2)$$

The next hopping term is

$$\sum_j \psi_B^{\dagger}(\mathbf{r}_j - \mathbf{a}) \psi_A(\mathbf{r}_j) = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{a}} c_{\mathbf{k}B}^{\dagger} c_{\mathbf{k}A}, \quad (4.4.3)$$

with Hermitian conjugate

$$\sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{a}} c_{\mathbf{k}A}^{\dagger} c_{\mathbf{k}B}. \quad (4.4.4)$$

The final hopping term is

$$\sum_j \psi_B^{\dagger}(\mathbf{r}_j - \mathbf{b}) \psi_A(\mathbf{r}_j) = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{b}} c_{\mathbf{k}B}^{\dagger} c_{\mathbf{k}A}, \quad (4.4.5)$$

with Hermitian conjugate

$$\sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{b}} c_{\mathbf{k}A}^{\dagger} c_{\mathbf{k}B}. \quad (4.4.6)$$

The number term is

$$\sum_i n_{Ai} n_{Bi} = \sum_{\mathbf{k}} c_{\mathbf{k}B}^{\dagger} c_{\mathbf{k}B} + c_{\mathbf{k}A}^{\dagger} c_{\mathbf{k}A}. \quad (4.4.7)$$

The full Hamiltonian can be written as

$$\hat{H} = \sum_{\mathbf{k}} \begin{pmatrix} c_{\mathbf{k}B}^{\dagger} & c_{\mathbf{k}A}^{\dagger} \end{pmatrix} \begin{pmatrix} \epsilon - \mu & \Delta \\ \Delta^* & \epsilon - \mu \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}B} \\ c_{\mathbf{k}A} \end{pmatrix}, \quad (4.4.8)$$

where

$$\Delta = -t(1 + e^{i\mathbf{k} \cdot \mathbf{a}} + e^{i\mathbf{k} \cdot \mathbf{b}}) \quad (4.4.9)$$

and

$$\Delta^* = -t(1 + e^{-i\mathbf{k} \cdot \mathbf{a}} + e^{-i\mathbf{k} \cdot \mathbf{b}}). \quad (4.4.10)$$

(c) It is easy to confirm that for

$$\mathbf{k} = \pm \mathbf{K} = \pm \begin{pmatrix} 0 \\ \frac{4\pi}{3a} \end{pmatrix}, \quad (4.4.11)$$

we have

$$\begin{cases} \mathbf{k} \cdot \mathbf{a} = \pm \frac{2\pi}{3} \\ \mathbf{k} \cdot \mathbf{b} = \mp \frac{2\pi}{3}. \end{cases} \quad (4.4.12)$$

Then

$$\Delta = -t(1 + 2 \cos(\frac{2\pi}{3})) = 0. \quad (4.4.13)$$

(d) Let $\mathbf{k} = \pm \mathbf{K} + \mathbf{p}$. For small $\|\mathbf{p}\|$,

$$\begin{aligned} \Delta(\mathbf{k}) &= -t \left(1 + e^{\pm \frac{2\pi}{3} i} e^{i\mathbf{p} \cdot \mathbf{a}} + e^{\mp \frac{2\pi}{3} i} e^{i\mathbf{p} \cdot \mathbf{b}} \right) \\ &= -t \left(1 + (-\frac{1}{2} \pm \frac{\sqrt{3}}{2} i)(1 + i\mathbf{p} \cdot \mathbf{a} + O(p^2)) + (-\frac{1}{2} \mp \frac{\sqrt{3}}{2} i)(1 + i\mathbf{p} \cdot \mathbf{b} + O(p^2)) \right) \\ &= -t \left(i\mathbf{p} \cdot \left(\frac{\mathbf{a} + \mathbf{b}}{2} \right) \pm \frac{\sqrt{3}}{2} \mathbf{p} \cdot (\mathbf{a} - \mathbf{b}) + O(p^2) \right) \\ &= \pm \tilde{c}(p_y \pm ip_x), \end{aligned} \quad (4.4.14)$$

where

$$\tilde{c} = \frac{\sqrt{3}}{2} at. \quad (4.4.15)$$

Therefore, the low-energy Hamiltonian is

$$\hat{H} = \sum_{\mathbf{p}} \begin{pmatrix} c_{\mathbf{k}B}^\dagger & c_{\mathbf{k}A}^\dagger \end{pmatrix} \begin{pmatrix} \epsilon - \mu & \pm \tilde{c}(p_y \pm ip_x) \\ \pm \tilde{c}(p_y \mp ip_x) & \epsilon - \mu \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}B} \\ c_{\mathbf{k}A} \end{pmatrix}. \quad (4.4.16)$$

Denoting

$$\psi_{\mathbf{p},+} = \begin{pmatrix} c_{\mathbf{p}+\mathbf{K},B} \\ c_{\mathbf{p}+\mathbf{K},A} \end{pmatrix} \quad \psi_{\mathbf{p},-} = \begin{pmatrix} c_{\mathbf{p}-\mathbf{K},A} \\ -c_{\mathbf{p}-\mathbf{K},B} \end{pmatrix}, \quad (4.4.17)$$

we can write

$$\hat{H} = \sum_{\mathbf{p}, \lambda \in \{\pm\}} \psi_{\mathbf{p},\lambda}^\dagger \begin{pmatrix} \epsilon - \mu & \tilde{c}(p_y + ip_x) \\ \tilde{c}(p_y - ip_x) & \epsilon - \mu \end{pmatrix} \psi_{\mathbf{p},\lambda}. \quad (4.4.18)$$

Now since

$$\boldsymbol{\sigma} \times \mathbf{p} = \sigma_x p_y - \sigma_y p_x = \begin{pmatrix} 0 & p_y + ip_x \\ p_y - ip_x & 0 \end{pmatrix}, \quad (4.4.19)$$

we can write

$$\hat{H} = \sum_{\mathbf{p}, \lambda \in \{\pm\}} \psi_{\mathbf{p},\lambda}^\dagger [\boldsymbol{\sigma} \times \mathbf{p} + (\epsilon - \mu) \mathbf{1}] \psi_{\mathbf{p},\lambda}. \quad (4.4.20)$$

When $\epsilon - \mu = 0$, the eigenvalue of the matrices is

$$\epsilon_{\mathbf{p}} = \pm |\Delta| = \pm \tilde{c} \|\mathbf{p}\|. \quad (4.4.21)$$

5 Green's Function

Exercise 5.1

(a) The Hamiltonian is

$$\hat{H}(t) = \underbrace{\omega_0 S_z}_{\hat{H}_0} + \underbrace{\omega_1 (S_x \cos \omega t + S_y \sin \omega t)}_{\hat{V}}, \quad (5.1.1)$$

where $\omega_0 = \gamma B_0$, $\omega_1 = \gamma B_1$, γ is the gyromagnetic ratio of the particle. The interaction Hamiltonian can be written as

$$\hat{V} = \frac{\omega_1}{2} (S_+ e^{-i\omega t} + S_- e^{i\omega t}). \quad (5.1.2)$$

The interaction-picture Schrödinger equation is

$$i \frac{d}{dt} |\psi_I(t)\rangle = \hat{H}_I(t) |\psi_I(t)\rangle, \quad (5.1.3)$$

where

$$\begin{aligned} \hat{H}_I(t) &= e^{i\hat{H}_0 t} \hat{V}(t) e^{-i\hat{H}_0 t} \\ &= e^{i\omega_0 S_z t} \hat{V} e^{-i\omega_0 S_z t}. \end{aligned} \quad (5.1.4)$$

Using $[S_z, S_{\pm}] = \pm S_{\pm}$, one can show that

$$e^{i\omega_0 S_z t} S_{\pm} e^{-i\omega_0 S_z t} = e^{\pm i\omega_0 t} S_{\pm}, \quad (5.1.5)$$

and hence

$$\begin{aligned} \hat{H}_I(t) &= \frac{\omega_1}{2} (S_+ e^{i(\omega_0 - \omega)t} + S_- e^{-i(\omega_0 - \omega)t}) \\ &= \frac{\omega_1}{2} (S_+ e^{i\delta t} + S_- e^{-i\delta t}), \end{aligned} \quad (5.1.6)$$

where we defined $\delta := \omega_0 - \omega$.

In the $(|\uparrow\rangle, |\downarrow\rangle)$ basis,

$$\hat{H}_I(t) = \frac{\omega_1}{2} \begin{pmatrix} 0 & e^{i\delta t} \\ e^{-i\delta t} & 0 \end{pmatrix}. \quad (5.1.7)$$

Let $|\psi_I(t)\rangle = (a(t), b(t))$, the interaction-picture Schrödinger equation is

$$i \frac{d}{dt} \begin{pmatrix} a(t) \\ b(t) \end{pmatrix} = \frac{\omega_1}{2} \begin{pmatrix} 0 & e^{i\delta t} \\ e^{-i\delta t} & 0 \end{pmatrix} \begin{pmatrix} a(t) \\ b(t) \end{pmatrix}. \quad (5.1.8)$$

(b) This is actually one of the few examples that the propagator is exactly solvable.

We can write

$$\begin{aligned} \hat{H}_I(t) &= \omega_1 (S_x \cos \delta t - S_y \sin \delta t) \\ &= e^{i\delta t S_z} (\omega_1 S_x) e^{-i\delta t S_z}. \end{aligned} \quad (5.1.9)$$

Inspired by the form of $\hat{H}_I$, we define the rotating frame by

$$|\phi(t)\rangle = e^{-i\delta t S_z} |\psi_I(t)\rangle. \quad (5.1.10)$$

Then

$$\frac{d}{dt} |\psi_I(t)\rangle = i\delta S_z e^{i\delta t S_z} |\phi\rangle + e^{i\delta t S_z} \frac{d}{dt} |\phi\rangle, \quad (5.1.11)$$

and

$$\hat{H}_I(t) |\psi_I(t)\rangle = e^{i\delta t S_z} \omega_1 S_x |\phi(t)\rangle, \quad (5.1.12)$$

and hence the interaction-picture Schrödinger equation (5.1.3) becomes

$$i \frac{d}{dt} |\phi(t)\rangle = (\omega_1 S_x + \delta S_z) |\phi(t)\rangle. \quad (5.1.13)$$

The effective Hamiltonian

$$\hat{H}_{\text{eff}} = \omega_1 S_x + \delta S_z \quad (5.1.14)$$

for the rotated state evolution is time-independent. And hence, this equation of motion is solvable

$$|\phi(t)\rangle = e^{-it\hat{H}_{\text{eff}}} |\phi(0)\rangle. \quad (5.1.15)$$

Transforming back into the lab frame,

$$|\psi_I(t)\rangle = e^{i\delta t S_z} e^{-it(\omega_1 S_x + \delta S_z)} |\psi_I(0)\rangle, \quad (5.1.16)$$

so the propagator is

$$U_I(t) = e^{i\delta t S_z} e^{-it(\omega_1 S_x + \delta S_z)}. \quad (5.1.17)$$

Defining the Rabi frequency $\Omega = \sqrt{\delta^2 + \omega_1^2}$,

$$\omega_1 S_x + \delta S_z = \frac{1}{2}(\delta \sigma_z + \omega_1 \sigma_x) = \frac{1}{2} \Omega \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}, \quad (5.1.18)$$

where

$$\hat{\mathbf{n}} = \frac{1}{\Omega}(\omega_1, 0, \delta). \quad (5.1.19)$$

Since $(\hat{\mathbf{n}} \cdot \boldsymbol{\sigma})^2 = 1$, so

$$e^{-i\frac{\Omega t}{2} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}} = \cos\left(\frac{\Omega t}{2}\right) 1 - i \sin\left(\frac{\Omega t}{2}\right) \hat{\mathbf{n}} \cdot \boldsymbol{\sigma}. \quad (5.1.20)$$

Therefore,

$$U_I(t) = e^{-\delta t \sigma_z / 2} \left[\cos\left(\frac{\Omega t}{2}\right) 1 - i \sin\left(\frac{\Omega t}{2}\right) \frac{\delta \sigma_z + \omega_1 \sigma_x}{\Omega} \right]. \quad (5.1.21)$$

In matrix form,

$$\begin{aligned} U_I(t) &= \begin{pmatrix} e^{i\delta t/2} & 0 \\ 0 & e^{-i\delta t/2} \end{pmatrix} \left[\cos\left(\frac{\Omega t}{2}\right) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \sin\left(\frac{\Omega t}{2}\right) \frac{1}{\Omega} \begin{pmatrix} \delta & \omega_1 \\ \omega_1 & -\delta \end{pmatrix} \right] \\ &= \begin{pmatrix} e^{i\delta t/2} \left(c - \frac{i\delta}{\Omega} s\right) & -ie^{i\delta t/2} \frac{\omega_1}{\Omega} s \\ -ie^{-i\delta t/2} \frac{\omega_1}{\Omega} s & e^{-i\delta t/2} \left(c + \frac{i\delta}{\Omega} s\right) \end{pmatrix}, \end{aligned} \quad (5.1.22)$$

where $c = \cos \frac{\Omega t}{2}$ and $s = \sin \frac{\Omega t}{2}$.

(c)

$$\begin{aligned} |\psi_I(t)\rangle &= U_I(t) |\psi_I(0)\rangle \\ &= \begin{pmatrix} e^{i\delta t/2} \left(c - \frac{i\delta}{\Omega} s\right) & -ie^{i\delta t/2} \frac{\omega_1}{\Omega} s \\ -ie^{-i\delta t/2} \frac{\omega_1}{\Omega} s & e^{-i\delta t/2} \left(c + \frac{i\delta}{\Omega} s\right) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} -ie^{i\delta t/2} \frac{\omega_1}{\Omega} s \\ e^{-i\delta t/2} \left(c + \frac{i\delta}{\Omega} s\right) \end{pmatrix}. \end{aligned} \quad (5.1.23)$$

The probability of remaining in $|\downarrow\rangle$ is

$$P(|\downarrow\rangle) = \left| e^{-i\delta t/2} \left(c + \frac{i\delta}{\Omega} s\right) \right|^2 = 1 - \frac{\omega_1^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right). \quad (5.1.24)$$